DOUBLING IN GENERALIZED POLYGONS: LARGE INCIDENCE STRUCTURES OF DIAMETER 3, 4 AND 6

DRAFT FOR DISCUSSION — AUTHORS TO BE SETTLED

ABSTRACT. Let $n_b(\Delta, D)$ be the largest number of elements (points plus lines) of an incidence structure in which every point lies on at most Δ lines, every line has at most Δ points, and any two elements are at distance at most D in the incidence graph. For $D \in \{3, 4, 6\}$ and $\Delta - 1$ a prime power the classical generalized D-gons of order $(\Delta - 1, \Delta - 1)$ attain the Moore bound. For the remaining degrees we adjoin new points and new lines to a classical generalized polygon. The main tool is Delorme's *doubling* of a set of elements, with levels, which we recast as an operation on incidence structures. We show that in a generalized n-gon, doubling a set of elements keeps the diameter at most n if and only if every connected component of that set is a tree of diameter at most $n - 2$. For $D = 3$ we add four *hats* to any projective plane of order $q \geq 3$ and obtain $2(q^2 + q + 6)$ elements at $\Delta = q + 2$; in $\mathrm{PG}(2, q)$ a correlation completes it to a regular structure. For $D = 4$ we show that two elliptic quadrics through a point x of $\mathrm{W}(q)$ form a weighted 2-ovoid of $\mathrm{AS}(q) = P(\mathrm{W}(q), x)$, q odd, which gives $(q+2)$-regular structures with $2q^2(q+2)$ elements; we also treat copies of lines ("shadows") and give a short proof that two orbits of a Singer subgroup form a 2-ovoid of $\mathrm{W}(q)$. For q even the quadrangle $\mathrm{Q}(4, q)$ has an ovoid and a spread, and doubling them gives $(q + 2)$-regular structures with $2(q^2 + 1)(q + 2)$ elements. For $D = 6$ we extend Delorme's doubling of the split Cayley hexagon $\mathrm{H}(q)$ along a Hermitian spread by an explicit set of $q^2(q^2 - 1)$ lines that meets the counting bound, for every odd q and every even $q \geq 4$ (for $q = 2$ by a set found by computer), and obtain $2(q+1)(q^4+q^2+1)+2q^4+4q^3+4q+10$ elements at $\Delta = q+2$, together with regular versions. The even case needs different hyperplanes and a different formula; it gives no new record in the range $\Delta \leq 16$.

1. INTRODUCTION

Let $\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$ be a finite incidence structure, and let $\Gamma(\mathcal{S})$ be its incidence graph. Every bipartite graph with a chosen bipartition is the incidence graph of an incidence structure. So the *bipartite degree/diameter problem* asks for the largest number of elements of an incidence structure with given maximum degree and given diameter. Write

$$n_b(\Delta, D) = \max\big\{\, |\mathcal{P}| + |\mathcal{L}| \;:\; \Delta(\mathcal{S}) \leq \Delta,\ \operatorname{diam}\mathcal{S} \leq D \,\big\},$$

where $\Delta(\mathcal{S})$ is the largest number of lines on a point or points on a line, and $\operatorname{diam}\mathcal{S}$ is the diameter of $\Gamma(\mathcal{S})$. Counting the elements at distance $0, 1, \ldots, D - 1$ from a flag gives the bipartite Moore bound

$$n_b(\Delta, D) \leq M_b(\Delta, D) = 2\big(1 + (\Delta - 1) + \cdots + (\Delta - 1)^{D-1}\big).$$

Equality holds exactly for the incidence graphs of generalized D-gons of order $(\Delta - 1, \Delta - 1)$. By the theorem of Feit and Higman [11] this forces $D \in \{3, 4, 6\}$ when $\Delta \geq 3$ (an octagon of order (s, s) would need $2s^2$ to be a square). Projective planes, generalized quadrangles and generalized hexagons of order (q, q) are known when q is a prime power, and for no other q. So the open cases with $D \in \{3, 4, 6\}$ are the degrees Δ for which $\Delta - 1$ is not a prime power. In the usual range $\Delta \leq 16$ these are $\Delta \in \{7, 11, 13, 15, 16\}$.

Our constructions all have the same shape. We start from a classical generalized polygon $\mathcal{S}$ of order (q, q) or $(q - 1, q + 1)$, with $q + 1 < \Delta$, and adjoin new points and new lines with prescribed

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incidences. The incidence graph appears only at the end, as the object whose order and diameter we count.

Results.

- *Doubling* (Section 3). We describe Delorme's doubling with levels [7, 8] as an operation $\mathcal{S} \mapsto \mathcal{S}_{\mathcal{A}}^{(m)}$ on incidence structures, prove Delorme's diameter theorem and its converse (Theorem 3.2), and show (Lemma 3.3 and Corollary 3.4) that in a generalized n-gon the doubling has diameter at most n if and only if each component of the doubled set is a tree of diameter at most $n-2$. This explains why diameter 3 needs extra "patch" incidences and why diameters 4 and 6 do not.
- *Diameter* 3 (Section 4). A fourth hat in any projective plane of order $q \geq 3$ (Theorem 4.2), completing the construction of Araujo-Pardo and López [2]: $n_b(q+2, 3) \geq 2(q^2+q+6)$, with a regular version in $\mathrm{PG}(2, q)$.
- *Diameter* 4 (Section 5). A weighted 2-ovoid of $\mathrm{AS}(q)$ formed by two elliptic quadrics (Theorem 5.6); clones and shadow lines in any generalized quadrangle (Theorem 5.4); a 2-ovoid of $\mathrm{W}(q)$ from a Singer subgroup (Proposition 5.8); regular structures of every degree $\Delta \geq q+1$ and every prime power q (Corollary 5.10; for even q it uses an ovoid and a spread of $\mathrm{Q}(4, q)$, see Section 7.1).
- *Diameter* 6 (Section 6). Delorme's doubling of $\mathrm{H}(q)$ along a Hermitian spread, extended by an explicit exact set of $q^2(q^2-1)$ "whisker" lines for every odd q (Theorem 6.5), and regular versions (Theorem 6.12).
- *Even* q (Section 7). In characteristic 2 the hyperplanes of Section 6 must avoid the nucleus of $\mathrm{Q}(6, q)$ and the whisker rule changes (Theorem 7.5); it gives an exact whisker set for every even $q \geq 4$, so that the bound of Corollary 6.9 and Theorem 6.12 hold for all such q (Corollary 7.6). For $q = 2$ the rule fails and we use an exact set found by computer, so the case $q = 2$ is computer-assisted. For $D = 4$ nothing new is needed.
- *Orders* (Section 8). Doubling gains at most about $2q^{D-2}$ elements at $\Delta = q+2$; the constructions reach this order of magnitude for $D = 4$ and attain the counting bound for $D = 6$.

Which q. Theorem 4.2 holds in every projective plane of order $q \geq 3$, Theorem 5.4 in every generalized quadrangle. Theorems 5.6 and 6.5, Corollaries 5.7 and 6.9, Proposition 5.8 and Theorem 6.12 are stated and proved for odd q in Sections 5 and 6. Corollary 5.10 holds for all prime powers q. For $D = 6$, Theorem 7.5 (even $q \geq 4$) and Corollary 7.6 (Corollary 6.9 and Theorem 6.12 for every prime power q, with $q = 2$ computer-assisted) are in Section 7. Table 3 uses odd q throughout; even q does not improve it.

2. Notation and preliminaries

2.1. **Incidence structures and distance.** An *incidence structure* $\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$ consists of disjoint sets $\mathcal{P}$ of *points* and $\mathcal{L}$ of *lines* and a symmetric relation $\mathrm{I} \subseteq (\mathcal{P} \times \mathcal{L}) \cup (\mathcal{L} \times \mathcal{P})$. Points and lines are *elements*. We do not assume that two points lie on at most one line: some of our structures have repeated lines. For a set $\mathcal{A}$ of elements, the *substructure induced on* $\mathcal{A}$ is $(\mathcal{A} \cap \mathcal{P}, \mathcal{A} \cap \mathcal{L}, \mathrm{I})$; its connected components are the components of the subgraph of $\Gamma(\mathcal{S})$ induced on $\mathcal{A}$. We say that $\mathcal{A}$ is a *tree* if this subgraph is a tree.

All distances are taken in $\Gamma(\mathcal{S})$. Two points are at even distance, a point and a line at odd distance. Two distinct points are at distance 2 exactly when they are collinear, and a point p and a line L are at distance 3 exactly when $p \not\mathrm{I} L$ and some point of L is collinear with p. An element is *isolated* if it is incident with nothing.

Lemma 2.1 (Diameter in geometric terms)**.** *Let $\mathcal{S}$ be an incidence structure without isolated elements and let $D \geq 2$.*

(i) *If D is even, then* $\operatorname{diam}\mathcal{S} \leq D$ *if and only if* $\mathrm{d}(p, L) \leq D-1$ *for every anti-flag (p, L).*

Symbol	Meaning
$\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$	incidence structure: points, lines, symmetric incidence relation
$\mathcal{S}^D = (\mathcal{L}, \mathcal{P}, \mathrm{I})$	the dual structure
$\Gamma(\mathcal{S})$	incidence graph: vertex set $\mathcal{P} \cup \mathcal{L}$, edges the flags
$\mathrm{d}(x, y)$	distance in $\Gamma(\mathcal{S})$; $\operatorname{diam}\mathcal{S} := \operatorname{diam}\Gamma(\mathcal{S})$
$\deg x$	number of elements incident with x (lines on a point; points on a line)
$\Delta(\mathcal{S})$	maximum of $\deg x$ over all elements
$x \sim y$	distinct points on a common line (collinear), or distinct lines on a common point (concurrent)
$x^\perp$	$\{x\} \cup \{y : y \sim x\}$
flag, anti-flag	incident, resp. non-incident, point–line pair
order (s, t)	every line has $s+1$ points, every point is on $t+1$ lines
opposite	at distance n in a generalized n-gon
$\mathrm{PG}(2, q)$; $(x{:}y{:}z)$, $[u{:}v{:}w]$	Desarguesian plane; points, lines; incidence $ux + vy + wz = 0$
$\mathrm{W}(q) = \mathrm{W}(3, q)$	symplectic quadrangle: points of $\mathrm{PG}(3, q)$, totally isotropic lines
$\mathrm{Q}(4, q)$, $\mathrm{Q}(6, q)$	parabolic quadrics; $\mathrm{Q}(4, q)$ is the dual of $\mathrm{W}(q)$
$\mathrm{AS}(q) = P(\mathrm{W}(q), x)$	Payne derivation of $\mathrm{W}(q)$ at x; order $(q-1, q+1)$; Ahrens–Szekeres quadrangle for q odd
$\mathrm{H}(q)$	split Cayley hexagon on $\mathrm{Q}(6, q)$, order (q, q)
$\mathcal{O}$; m-ovoid; weighted m-ovoid	ovoid; point set meeting every line in m points; weight $w\colon \mathcal{P} \to \mathbb{Z}_{\geq 0}$ with $\sum_{p \mathrm{I} L} w(p) = m$ for every line L
spread; m-cover	duals of ovoid and of (weighted) m-ovoid
$\mathcal{A} = (\mathcal{A}_0 \supseteq \mathcal{A}_1 \supseteq \cdots)$	a filtration of a set of elements; $\lambda(x)$ the level of x
$\mathcal{S}_{\mathcal{A}}^{(m)}$, x^w	m-fold doubling of $\mathcal{S}$ along $\mathcal{A}$; the version of x indexed by the word w
$n_b(\Delta, D)$, $M_b(\Delta, D)$	largest order; bipartite Moore bound

TABLE 1. Notation.

(ii) *If D is odd, then* $\operatorname{diam}\mathcal{S} \leq D$ *if and only if any two points are at distance at most $D-1$ and any two lines are at distance at most $D-1$.*

In particular:

- $\operatorname{diam}\mathcal{S} \leq 3$ *if and only if any two points are collinear and any two lines are concurrent;*
- $\operatorname{diam}\mathcal{S} \leq 4$ *if and only if for every anti-flag (p, L) some point of L is collinear with p;*
- $\operatorname{diam}\mathcal{S} \leq 6$ *if and only if for every anti-flag (p, L) some point x I L and some point y satisfy $p \in y^\perp$ and $x \in y^\perp$.*

Proof. (i) Let $p \neq p'$ be points. Choose a line L I p'. If p I L then $\mathrm{d}(p, p') = 2$; otherwise $\mathrm{d}(p, p') \leq \mathrm{d}(p, L) + 1 \leq D$. Two lines are handled in the same way, choosing a point on one of them. A flag is at distance 1. (ii) Let (p, L) be an anti-flag and choose p' I L. Then $p' \neq p$ and $\mathrm{d}(p, L) \leq \mathrm{d}(p, p') + 1 \leq D$. The converse directions of (i) and (ii) hold by parity. The three special cases are the translations of $\mathrm{d} \leq 2$, $\mathrm{d} \leq 3$ and $\mathrm{d} \leq 5$. □

2.2. **Generalized polygons.** Following Tits and [18], a *generalized n-gon* ($n \geq 3$) is an incidence structure whose incidence graph has diameter n and girth $2n$. It has *order* (s, t) if every line has $s+1$ points and every point is on $t+1$ lines; it is *thick* if $s, t \geq 2$. Elements at distance n are *opposite*. We use two consequences of the girth.

(G1) If $\mathrm{d}(x, y) < n$, there is a unique path of length $\mathrm{d}(x, y)$ from x to y.

(G2) A generalized quadrangle ($n = 4$) of order (s, t) has $(s+1)(st+1)$ points and $(t+1)(st+1)$ lines; for an anti-flag (p, L) there is a unique point of L collinear with p. A generalized hexagon of order (s, t) has $(s+1)(s^2t^2+st+1)$ points and $(t+1)(s^2t^2+st+1)$ lines.

A generalized n-gon of order (q, q) has $2(1 + q + \cdots + q^{n-1}) = M_b(q+1, n)$ elements. By Feit and Higman [11], finite thick generalized n-gons exist only for $n \in \{3, 4, 6, 8\}$, and for $n = 8$ the number $2st$ is a square.

2.3. **The classical examples.** Let q be a prime power and $\mathbb{F} = \mathrm{GF}(q)$.

The plane $\mathrm{PG}(2, q)$ [12]. Points $(x{:}y{:}z)$ and lines $[u{:}v{:}w]$ are nonzero vectors of $\mathbb{F}^3$ up to scalars, and $(x{:}y{:}z)$ I $[u{:}v{:}w]$ if and only if $ux + vy + wz = 0$. It is a generalized 3-gon (projective plane) of order (q, q). Non-Desarguesian projective planes also exist for many q; Section 4 works in any projective plane.

The quadrangles $\mathrm{W}(q)$ *and* $\mathrm{Q}(4, q)$ [14, 3.1.1, 3.2.1]. $\mathrm{W}(q) = \mathrm{W}(3, q)$ has as points all points of $\mathrm{PG}(3, q)$ and as lines the totally isotropic lines of a symplectic polarity; $\mathrm{Q}(4, q)$ has as points and lines those of a non-singular quadric of $\mathrm{PG}(4, q)$. Both have order (q, q), and $\mathrm{Q}(4, q)$ is isomorphic to the dual of $\mathrm{W}(q)$. An *ovoid* of a generalized quadrangle is a set of points meeting every line in exactly one point; a *spread* is the dual notion. A *partial ovoid* meets every line in at most one point. $\mathrm{Q}(4, q)$ has ovoids for every q (the elliptic quadric sections $\mathrm{Q}^-(3, q)$); $\mathrm{W}(q)$ has ovoids if and only if q is even, so $\mathrm{Q}(4, q)$ has spreads if and only if q is even [14, 3.4.1]. $\mathrm{W}(q)$ always has spreads, for instance regular spreads of $\mathrm{PG}(3, q)$.

m-ovoids and weights. An *m-ovoid* is a set of points meeting every line in exactly m points [3]. A *weighted m-ovoid* is a function $w\colon \mathcal{P} \to \mathbb{Z}_{\geq 0}$ with $\sum_{p \mathrm{I} L} w(p) = m$ for every line L; equivalently a multiset of points meeting every line in m points counted with multiplicity. Dually, a *(weighted) m-cover* is a set (multiset) of lines with exactly m members through every point. By duality, a weighted m-ovoid of $\mathrm{W}(q)$ is a weighted m-cover of $\mathrm{Q}(4, q)$.

The quadrangle $\mathrm{AS}(q)$. Every point x of $\mathrm{W}(q)$ is regular. Payne's derivation $P(\mathrm{W}(q), x)$ [13], [14, 3.1.4] has as points the points of $\mathrm{PG}(3, q)$ not in $x^\perp$, and as lines (i) the totally isotropic lines not through x, each with its point in $x^\perp$ removed, and (ii) the lines of $\mathrm{PG}(3, q)$ through x not contained in $x^\perp$ (the hyperbolic lines $\{x, y\}^{\perp\perp}$), each with x removed. It is a generalized quadrangle of order $(q-1, q+1)$, and for q odd it is isomorphic to the Ahrens–Szekeres quadrangle $\mathrm{AS}(q)$ [1], [14, 3.1.5, 3.2.6]. The lines of type (ii) form a spread of $\mathrm{AS}(q)$.

The hexagon $\mathrm{H}(q)$ [17], [18, §2.4]. Let $\mathrm{Q}(6, q)$ be the quadric $X_0X_4 + X_1X_5 + X_2X_6 = X_3^2$ of $\mathrm{PG}(6, q)$. The points of $\mathrm{H}(q)$ are all points of $\mathrm{Q}(6, q)$; the lines are the lines of $\mathrm{Q}(6, q)$ whose Grassmann coordinates $p_{ij} = x_iy_j - x_jy_i$ satisfy Tits' equations

$$p_{12} = p_{34}, \quad p_{54} = p_{32}, \quad p_{20} = p_{35}, \quad p_{65} = p_{30}, \quad p_{01} = p_{36}, \quad p_{46} = p_{31}.$$

$\mathrm{H}(q)$ is a generalized hexagon of order (q, q). For a point x, the conditions that a point y lies on a line of $\mathrm{H}(q)$ through x are linear in y; the $q+1$ lines of $\mathrm{H}(q)$ through x are the lines through x of a totally singular plane π_x of $\mathrm{Q}(6, q)$ (the hexagon is *flat*). A *Hermitian spread* of $\mathrm{H}(q)$ is the set of lines of $\mathrm{H}(q)$ contained in a hyperplane Π for which $\Pi \cap \mathrm{Q}(6, q)$ is an elliptic quadric $\mathrm{Q}^-(5, q)$ [16], [18]; Lemma 6.1 proves the properties we use.

3. Doubling with levels

Delorme and Farhi [9] observed that a *duplicate* of an element (a new element incident with exactly the same elements) does not increase the diameter. Delorme [7, 8] refined this into a doubling of a whole set of elements, with levels, and used it to build the largest bipartite graphs then known of diameters 4 and 6. We recast it as an operation on incidence structures.[1]

[1]Note for co-authors: the full texts of Delorme's two 1985 papers [7, 8] have not been read by the authors of this draft. Our earlier notes attribute the doubling with levels and the bipartite constructions for $D = 4, 6$ to [7],

3.1. **The operation.** Let $\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$ be an incidence structure, $m \geq 2$ an integer, and

$$\mathcal{A} = (\mathcal{A}_0 \supseteq \mathcal{A}_1 \supseteq \cdots \supseteq \mathcal{A}_r \supseteq \mathcal{A}_{r+1} = \emptyset)$$

a filtration of a set $\mathcal{A}_0$ of elements (points and lines may be mixed). The *level* of an element x is $\lambda(x) = \max\{k : x \in \mathcal{A}_k\}$, with $\lambda(x) = -1$ if $x \notin \mathcal{A}_0$. Let $[m] = \{1, \ldots, m\}$ and write $u \preceq w$ if the word u is a prefix of the word w; two words are *comparable* if one is a prefix of the other.

Definition 3.1 (Doubling)**.** The *m-fold doubling of $\mathcal{S}$ along $\mathcal{A}$* is the incidence structure $\mathcal{S}_{\mathcal{A}}^{(m)}$ whose elements are the *versions*

$$x^w, \qquad x \in \mathcal{P} \cup \mathcal{L}, \quad w \in [m]^{\lambda(x)+1},$$

where x^w is a point if x is a point and a line if x is a line, and in which

$$x^w \,\mathrm{I}\, y^u \iff x \,\mathrm{I}\, y \text{ and } w, u \text{ are comparable.}$$

An element of level -1 has the single version $x^{\varnothing}$, which we identify with x. For $m = 2$ we speak of *doubling*, for $m = 3$ of *tripling*.

For a single level ($r = 0$, $\mathcal{A} = \mathcal{A}_0$) and $m = 2$, write $x = x^1$ and $x' = x^2$ for $x \in \mathcal{A}$. Then $\mathcal{S}_{\mathcal{A}}^{(2)}$ is obtained from $\mathcal{S}$ by adjoining, for each $x \in \mathcal{A}$, a new element x' of the same type, incident with

$$y' \text{ for every } y \in \mathcal{A} \text{ with } y \,\mathrm{I}\, x, \qquad \text{and} \qquad y \text{ for every } y \notin \mathcal{A} \text{ with } y \,\mathrm{I}\, x.$$

If $\mathcal{A}$ contains no flag, x' is a duplicate of x in the sense of [9]. In general x' is *not* a duplicate: inside $\mathcal{A}$ the copies are incident with copies, outside $\mathcal{A}$ with originals.

The map $x^w \mapsto x$ preserves incidence, so

$$\mathrm{d}(x^w, y^u) \geq \mathrm{d}(x, y) \qquad \text{for all versions.} \tag{1}$$

In particular two opposite elements of level -1 stay at distance n.

3.2. **Delorme's theorem.** Consider the following conditions on $\mathcal{A}$, where $D \geq 2$.

- (L0) Every element outside $\mathcal{A}_0$ is incident with at most one element of $\mathcal{A}_0$.
- (L1) For every $i \geq 0$: every element incident with an element of $\mathcal{A}_{i+1}$ lies in $\mathcal{A}_i$, and every element of $\mathcal{A}_i \setminus \mathcal{A}_{i+1}$ is incident with at most one element of $\mathcal{A}_{i+1}$.
- (C_D) For any two elements x, y (possibly $x = y$) in the same connected component of the substructure induced on $\mathcal{A}_0$, there is an element $z \notin \mathcal{A}_0$ with $\mathrm{d}(x, z) + \mathrm{d}(z, y) \leq D$.

Theorem 3.2 (Delorme [7, 8])**.** *Let $\mathcal{S}$ be an incidence structure of diameter at most D, $D \geq 2$, and let $m \geq 2$.*

(a) *If $\mathcal{A}$ satisfies* (C_D)*, then $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most D.*

(b) *If $\mathcal{A}$ satisfies* (L0) *and* (L1)*, then every version x^w has $\deg x^w \leq \deg x + m - 1$, with equality only if x is incident with an element of level $\lambda(x)+1$. In particular $\Delta(\mathcal{S}_{\mathcal{A}}^{(m)}) \leq \Delta(\mathcal{S}) + m - 1$.*

(c) *The numbers of points and of lines of $\mathcal{S}_{\mathcal{A}}^{(m)}$ are*

$$|\mathcal{P}| + (m-1)\sum_{k \geq 0} m^k |\mathcal{A}_k \cap \mathcal{P}| \qquad \text{and} \qquad |\mathcal{L}| + (m-1)\sum_{k \geq 0} m^k |\mathcal{A}_k \cap \mathcal{L}|.$$

(d) *Conversely, if $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most D, then* (C_D) *holds; indeed its conclusion holds for every pair $x, y \in \mathcal{A}_0$.*

We include a proof.

Sections 7–8; the abstract of [8] also describes bipartite constructions. Please check which paper contains which construction; no section numbers are claimed here.

Proof. (c) An element of level k has m^{k+1} versions, and $m^{k+1} - 1 = (m-1)(1 + m + \cdots + m^k)$. Summing over the levels gives the formula.

(b) Let $\lambda(x) = k$ and $|w| = k+1$. A neighbor y of x with $\lambda(y) = j \le k$ has exactly one version incident with x^w (the one indexed by the prefix of w of length $j+1$), and a neighbor with $\lambda(y) = j > k$ has m^{j-k} such versions. If $k = -1$, (L0) says that x has at most one neighbor in $\mathcal{A}_0$, and that neighbor has level 0, since a neighbor in $\mathcal{A}_1$ would put x in $\mathcal{A}_0$ by (L1). If $k \ge 0$, (L1) says that x has at most one neighbor in $\mathcal{A}_{k+1}$, and none in $\mathcal{A}_{k+2}$, since that would put x in $\mathcal{A}_{k+1}$. So $\deg x^w \le \deg x - 1 + m$, with equality only when the higher neighbor exists.

(a) For an infinite word ω over $[m]$, the *sheet* Σ_ω is the set of versions x^w with $w \preceq \omega$. Every element has exactly one version in Σ_ω, and any two versions in Σ_ω have comparable words, so $x^w \mapsto x$ is an isomorphism from the substructure on Σ_ω onto $\mathcal{S}$. Two versions lie in a common sheet if and only if their words are comparable, and then they are at distance at most D.

Let x^w, y^u have incomparable words. Then both words are non-empty, so $x, y \in \mathcal{A}_0$. If x and y lie in different components of the substructure on $\mathcal{A}_0$, a shortest path $x = z_0, z_1, \ldots, z_k = y$ in $\Gamma(\mathcal{S})$, $k \le D$, contains some $z_i \notin \mathcal{A}_0$. The element z_i has a single version, which lies in every sheet. Lift $z_0, \ldots, z_i$ into a sheet containing x^w and $z_i, \ldots, z_k$ into a sheet containing y^u; this is a path of length k from x^w to y^u. If x and y lie in the same component (this includes $x = y$), (C_D) gives $z \notin \mathcal{A}_0$ with $\mathrm{d}(x,z) + \mathrm{d}(z,y) \le D$, and the same lifting gives a path of length at most D.

(d) Let $x, y \in \mathcal{A}_0$ (possibly $x = y$). Both have versions with non-empty words, so choose x^w and y^u with w beginning with the letter 1 and u with the letter 2. Take a path of length at most D from x^w to y^u in $\mathcal{S}_{\mathcal{A}}^{(m)}$. Consecutive versions on it have comparable words, and two comparable non-empty words begin with the same letter. So some version on the path has the empty word; it is the unique version of an element $z \notin \mathcal{A}_0$. The map $x^w \mapsto x$ preserves incidence, so the path projects to a walk from x through z to y of length at most D, whence $\mathrm{d}(x,z) + \mathrm{d}(z,y) \le D$. □

3.3. **The hat lemma.** In a generalized polygon, condition (C_D) has a clean geometric form.

Lemma 3.3 (Hat lemma)**.** *Let $\mathcal{S}$ be a generalized n-gon of order (s,t) with $s, t \ge 1$, and let $\mathcal{A}_0$ be a set of elements. Then (C_n) holds if and only if every connected component of the substructure induced on $\mathcal{A}_0$ is a tree of diameter at most $n-2$.*

Proof. Suppose some component B contains a path $x = z_0, z_1, \ldots, z_{n-1} = y$ of length $n-1$. (It does if B has diameter at least $n-1$; it does if B contains a cycle, because cycles of $\Gamma(\mathcal{S})$ have length at least $2n$.) By the girth, this is the unique path of length at most $n-1$ from x to y, so $\mathrm{d}(x,y) = n-1$. If $\mathrm{d}(x,z) + \mathrm{d}(z,y) \le n$, then by parity $\mathrm{d}(x,z) + \mathrm{d}(z,y) = n-1$, so z lies on a shortest path from x to y, that is $z = z_i \in \mathcal{A}_0$. So ($\mathrm{C}_n$) fails.

Conversely, let B be a tree of diameter at most $n-2$, let $x, y \in B$, and let P be the path from x to y in B, of length d. The component B is not all of $\Gamma(\mathcal{S})$, which contains cycles, so some element of B has a neighbor outside $\mathcal{A}_0$. Let k be the least distance in B from P to such an element u, and let $z \notin \mathcal{A}_0$ be a neighbor of u. Then $\mathrm{d}(x,z) + \mathrm{d}(z,y) \le d + 2k + 2$. If $k = 0$ this is at most n. If $k \ge 1$, every element of B at distance less than k from P has all its neighbors in B, and every element has at least two neighbors. Walking k steps away from P from x, and k steps away from P from y (along two different neighbors if $x = y$), we obtain a path of length $d + 2k$ in the tree B. Hence $d + 2k \le n-2$, and again $\mathrm{d}(x,z) + \mathrm{d}(z,y) \le n$. □

Corollary 3.4. *Let $\mathcal{S}$ be a generalized n-gon of order (s,t) with $s, t \ge 1$, let $\mathcal{A}$ be a filtration of a set $\mathcal{A}_0$ of elements, and let $m \ge 2$. Then $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most n if and only if every connected component of the substructure induced on $\mathcal{A}_0$ is a tree of diameter at most $n-2$.*

Proof. By Theorem 3.2(a) and (d), the diameter is at most n if and only if (C_n) holds; apply Lemma 3.3. □

In geometric terms, the components allowed by Lemma 3.3 are:

- $n = 3$ (projective planes): single elements and flags;
- $n = 4$ (quadrangles): single elements, flags, and *stars*: a point with some of the lines on it, or a line with some of its points;
- $n = 6$ (hexagons): trees of diameter at most 4, for instance an element x, some elements incident with x, and some elements incident with those.

3.4. **Two consequences.**

Proposition 3.5 (Diameter 3 needs patches)**.** *Let $\mathcal{S}$ be a projective plane and let $\mathcal{A}$ satisfy* (L0)*,* (L1) *and* (C_3)*. Then $\mathcal{A}_0$ consists of at most one point and at most one line, and $\mathcal{A}_1 = \emptyset$.*

Proof. Let $x \neq y$ be points of $\mathcal{A}_0$. If the line xy is not in $\mathcal{A}_0$, it is incident with two elements of $\mathcal{A}_0$, against (L0). If it is in $\mathcal{A}_0$, then x, xy, y lie in one component, which is then not a single element or a flag, against Lemma 3.3. The same holds for lines. If $\mathcal{A}_1$ contained an element, its $q + 1 \geq 3$ neighbors would lie in $\mathcal{A}_0$ by (L1). □

So a pure doubling of a projective plane gains at most two elements. The constructions of Section 4 double flags and then modify a bounded number of incidences (*patches*).

Proposition 3.6 (Counting bound)**.** *Let $\mathcal{S}$ be a generalized n-gon of order (q, q) with N points and N lines, and let $\mathcal{A}_0$ satisfy* (L0) *and* (C_n)*. Let c be the number of components of the substructure on $\mathcal{A}_0$. Then*

$$q\,|\mathcal{A}_0| + 2c \leq 2N,$$

with equality if and only if every element outside $\mathcal{A}_0$ is incident with exactly one element of $\mathcal{A}_0$.

Proof. By Lemma 3.3 the substructure on $\mathcal{A}_0$ is a forest with $|\mathcal{A}_0| - c$ flags. So the number of flags with exactly one element in $\mathcal{A}_0$ is $(q + 1)|\mathcal{A}_0| - 2(|\mathcal{A}_0| - c)$. By (L0) it is at most $2N - |\mathcal{A}_0|$, with equality as stated. □

Since $N \sim q^{n-1}$, a doubling along $\mathcal{A}$ at $\Delta = q + 2$ has $|\mathcal{A}_0| < 2N/q \sim 2q^{n-2}$; the higher levels contribute lower-order terms. Section 8 compares this with the Moore bound.

4. Diameter 3: four hats in a projective plane

By Lemma 2.1, an incidence structure without isolated elements has diameter at most 3 exactly when any two points are collinear and any two lines are concurrent. Araujo-Pardo and López [2] add three *hats* to $\mathrm{PG}(2, q)$ and obtain $n_b(q + 2, 3) \geq 2(q^2 + q + 5)$. We add a fourth hat.

4.1. **The frame.** Let Π be a projective plane of order $q \geq 3$. Fix an anti-flag (A, ℓ), and four distinct points of ℓ, labeled X_s with s in a four-element index set $S = \{0, 1, a, c\}$. Put

$$T = S \setminus \{a\} = \{0, 1, c\}, \qquad \lambda_s = AX_s \ (s \in S),$$

and let σ be a 3-cycle on T; for definiteness $\sigma = (0\ c\ 1)$, that is $\sigma(0) = c$, $\sigma(c) = 1$, $\sigma(1) = 0$. Let

$$\ell^* = \ell \setminus \{X_t : t \in T\} \qquad \text{and} \qquad \Lambda^* = \{\text{lines on } A\} \setminus \{\lambda_t : t \in T\}.$$

So ℓ^* consists of $q - 2$ points of ℓ, among them X_a, and Λ^* of $q - 2$ lines on A, among them λ_a.

Definition 4.1 (Four hats)**.** The incidence structure $\mathcal{S} = \mathcal{S}(\Pi; A, \ell, (X_s)_{s\in S}, \sigma)$ has as points the points of Π and five new points Q, H_s ($s \in S$), and as lines the lines of Π and five new lines M, K_s ($s \in S$). Its incidences are exactly the following.

(H1) The incidences of Π, except the four flags (X_s, λ_s), $s \in S$; and the three *rotated* flags $(X_t, \lambda_{\sigma(t)})$, $t \in T$:

$$X_0 \mathrel{\mathrm{I}} \lambda_c, \qquad X_c \mathrel{\mathrm{I}} \lambda_1, \qquad X_1 \mathrel{\mathrm{I}} \lambda_0.$$

(H2) For $s \in S$, the hat point H_s lies on every line of Π through X_s other than ℓ.

(H3) For $s \in S$, the hat line K_s contains every point of λ_s other than A.

(H4) M contains the points of ℓ^* and the four hat points H_s.
(H5) Q lies on the lines of Λ^* and on the four hat lines K_s.
(H6) The *links*: $A \mathrel{\mathrm{I}} K_a$, $H_a \mathrel{\mathrm{I}} \ell$, and $H_t \mathrel{\mathrm{I}} K_{\sigma(t)}$ for $t \in T$:

$$H_0 \mathrel{\mathrm{I}} K_c, \qquad H_c \mathrel{\mathrm{I}} K_1, \qquad H_1 \mathrel{\mathrm{I}} K_0.$$

The hat pair (H_s, K_s) is a copy of the flag (X_s, λ_s): H_s lies on the lines through X_s and K_s contains the points of λ_s, with the flag "crossed" ($H_s \mathrel{\mathrm{I}} \lambda_s$ and $X_s \mathrel{\mathrm{I}} K_s$). The pair (Q, M) is a copy of the anti-flag (A, ℓ); see Remark 4.10.

Theorem 4.2. *Let Π be a projective plane of order $q \geq 3$. The incidence structure $\mathcal{S}$ of Definition 4.1 has q^2+q+6 points and q^2+q+6 lines, every element has degree $q+1$ or $q+2$, and $\mathcal{S}$ has diameter 3. Hence $n_b(q+2, 3) \geq 2(q^2+q+6)$ for every prime power $q \geq 3$.*

The proof occupies Lemmas 4.3–4.5.

Lemma 4.3 (Degrees)**.** *In $\mathcal{S}$:*

(i) *A, every point of ℓ, Q and every H_s are on exactly $q+2$ lines;*
(ii) *a point $p \notin \ell$, $p \neq A$, of Π is on $q+2$ lines if $Ap = \lambda_s$ for some $s \in S$, and on $q+1$ lines otherwise;*
(iii) *dually, ℓ, every line on A, M and every K_s have exactly $q+2$ points, and a line $L \neq \ell$ of Π not on A has $q+2$ points if $L \cap \ell = X_s$ for some $s \in S$, and $q+1$ points otherwise.*

In particular there are $(q-1)(q-3)$ points and $(q-1)(q-3)$ lines of degree $q+1$.

Proof. We count lines on each point: the number in Π, minus removed flags, plus added flags. A: its flags in Π are untouched, and it gains K_a (H6); it is not on any other K_s by H3. $X \in \ell^*$, $X \neq X_a$: gains M. X_a: loses λ_a, gains M and K_a (H3, as $X_a \neq A$ is on λ_a). X_t, $t \in T$: loses λ_t, gains $\lambda_{\sigma(t)}$ and K_t; it is not on M. A point $p \notin \ell$, $p \neq A$: it is on exactly one line on A, namely Ap; it loses nothing and gains K_s exactly when $Ap = \lambda_s$. Q: $|\Lambda^*| + 4 = q+2$ lines. H_s: q lines of Π, M, and one link (H6): ℓ if $s = a$, $K_{\sigma(s)}$ if $s \in T$. The statement for lines is the dual count (or follows from Lemma 4.5). Finally, the points of degree $q+1$ are the $q-1$ points other than A and X on each of the $q-3$ lines AX with $X \in \ell \setminus \{X_s : s \in S\}$. □

Lemma 4.4 (Points)**.** *Any two distinct points of $\mathcal{S}$ are collinear in $\mathcal{S}$.*

Proof. The only flags of Π missing from $\mathcal{S}$ are (X_s, λ_s), $s \in S$. Let $u \neq v$ be points of $\mathcal{S}$.

(P1) u, v in Π. Let $L = uv$ in Π. If both flags survive, L works. Otherwise, say, $u = X_s$ and $L = \lambda_s$. If $v \neq A$, both u and v lie on K_s by H3. If $v = A$ and $s = a$, use K_a (H3, H6). If $v = A$ and $s \in T$, use $\lambda_{\sigma(s)}$, which contains A and, by the rotation, X_s.

(P2) Q and a point u of Π. If $u = A$, use $\lambda_a \in \Lambda^*$. If $u \in \ell$ and $u \neq X_s$ for all s, use $Au \in \Lambda^*$. If $u = X_s$, use K_s. If $u \notin \ell$, $u \neq A$, let $X = Au \cap \ell$: if $Au \in \Lambda^*$ use Au; otherwise $X = X_t$ with $t \in T$ and u, Q lie on K_t.

(P3) H_s and a point u of Π. If $u \notin \ell$ (this includes $u = A$), the line uX_s of Π is not ℓ, contains u in $\mathcal{S}$ (as $u \notin \ell$) and contains H_s by H2. If $u = X_s$, use any line through X_s other than ℓ and λ_s. If $u \in \ell^*$, $u \neq X_s$, use M. Otherwise $u = X_t$ with $t \in T \setminus \{s\}$. If $s = a$, use ℓ (H6). If $s \in T$, then $t \in \{\sigma(s), \sigma^{-1}(s)\}$ because σ is a 3-cycle. If $t = \sigma(s)$, use $K_{\sigma(s)}$, which contains $X_{\sigma(s)}$ (H3) and H_s (H6). If $t = \sigma^{-1}(s)$, use λ_s, which contains $X_{\sigma^{-1}(s)}$ (rotation) and H_s (H2).

(P4) Two new points. H_s and $H_{s'}$: M. Q and H_a: λ_a ($\lambda_a \in \Lambda^*$, and $\lambda_a \neq \ell$ is a line through X_a). *Q and H_t, $t \in T$: $K_{\sigma(t)}$.* □

Lemma 4.5 (Duality)**.** *Let Π^D be the dual plane. The dual of $\mathcal{S}(\Pi; A, \ell, (X_s), \sigma)$ is isomorphic to $\mathcal{S}(\Pi^D; \ell, A, (\lambda_s), \sigma^{-1})$, by the identity on Π and*

$$H_s \leftrightarrow K_s, \qquad Q \leftrightarrow M.$$

Consequently any two distinct lines of $\mathcal{S}$ are concurrent.

Proof. In Π^D the point ℓ is off the line A, the points λ_s lie on the line A, and the line joining the points ℓ and λ_s is X_s. Read the six rules of Definition 4.1 in Π^D with these data and with σ^{-1}. Rule H1 removes the flags (λ_s, X_s) and adds λ_t I $X_{\sigma^{-1}(t)}$, which is the rotation X_u I $\lambda_{\sigma(u)}$ of $\mathcal{S}$. Rule H2 gives a new element incident with every element of Π incident with λ_s other than A: this is K_s. Rule H3 gives a new element incident with every element incident with X_s other than ℓ: this is H_s. Rule H4 gives a new element incident with the lines on A other than λ_t ($t \in T$) and with all K_s: this is Q; likewise H5 gives M. Rule H6 gives ℓ I H_a, K_a I A and K_t I $H_{\sigma^{-1}(t)}$, which are the links of $\mathcal{S}$. Since Π^D is a projective plane of order q and σ^{-1} is a 3-cycle on T, Lemma 4.4 for $\mathcal{S}(\Pi^D; \ell, A, (\lambda_s), \sigma^{-1})$ says that any two lines of $\mathcal{S}$ are concurrent. □

Proof of Theorem 4.2. The counts are Lemma 4.3; no element is isolated. By Lemmas 4.4, 4.5 and 2.1, $\operatorname{diam}\mathcal{S} \le 3$, and $\operatorname{diam}\mathcal{S} = 3$ because $A \not| \ell$. □

4.2. **Coordinates and a regular completion.** Now let $\Pi = \mathrm{PG}(2, q)$ with $q \ge 4$. Take

$$\ell = [1{:}0{:}0], \quad A = (1{:}0{:}0), \quad P = (0{:}1{:}0), \quad \lambda_\infty = [0{:}0{:}1], \qquad X_t = (0{:}t{:}1), \quad \lambda_t = [0{:}1{:}{-}t] \quad (t \in \mathbb{F}),$$

so that $\ell = \{P\} \cup \{X_t\}$, the lines on A are λ_∞ and the λ_t, $\lambda_t \cap \ell = X_t$ and $\lambda_\infty \cap \ell = P$. Let $a, c \in \mathbb{F} \setminus \{0, 1\}$ be distinct, and use the labels $S = \{0, 1, a, c\} \subseteq \mathbb{F}$ for the points X_0, X_1, X_a, X_c. Then $\ell^* = \{P\} \cup \{X_t : t \notin T\}$ and $\Lambda^* = \{\lambda_\infty\} \cup \{\lambda_t : t \notin T\}$, and the rules read:

- H_s ($s \in S$) lies on $[w{:}1{:}{-}s]$ for every $w \in \mathbb{F}$ (these are the lines on X_s other than ℓ, with λ_s at $w = 0$);
- K_s ($s \in S$) contains $(x{:}s{:}1)$ for every $x \in \mathbb{F}$ (the points of λ_s other than A, with X_s at $x = 0$);
- M contains P, X_t ($t \notin T$) and the H_s; Q lies on λ_∞, λ_t ($t \notin T$) and the K_s;
- rotation and links as in H1 and H6.

By Lemma 4.3 the elements of degree $q+1$ are the points $(x{:}y{:}1)$ ($x \ne 0$, $y \notin S$) and $(x{:}1{:}0)$ ($x \ne 0$), and the lines $[u{:}1{:}w]$ ($u \ne 0$, $-w \notin S$) and $[1{:}0{:}w]$ ($w \ne 0$).

Lemma 4.6 (A correlation)**.** *The map* $\delta\colon (x{:}y{:}z) \mapsto [x{:}z{:}{-}y]$, $[u{:}v{:}w] \mapsto (u{:}w{:}{-}v)$ *is a correlation of* $\mathrm{PG}(2, q)$ *with*

$$\delta(A) = \ell, \quad \delta(X_t) = \lambda_t, \quad \delta(P) = \lambda_\infty,$$

and a point p lies on $\delta(p)$ if and only if $p \in \ell$. It maps the points of degree $q+1$ bijectively onto the lines of degree $q+1$.

Proof. $ux + wz + (-v)(-y) = ux + vy + wz$, so δ preserves incidence; it is clearly bijective. $(x{:}y{:}z)$ I $[x{:}z{:}{-}y]$ iff $x^2 + yz - zy = x^2 = 0$. Finally $\delta(x{:}y{:}1) = [x{:}1{:}{-}y]$ with $x \ne 0$, $y \notin S$, and $\delta(x{:}1{:}0) = [x{:}0{:}{-}1] = [1{:}0{:}{-}1/x]$. □

Theorem 4.7. *Let $q \ge 4$ be a prime power and a, c distinct elements of $\mathbb{F} \setminus \{0, 1\}$. Adjoining to $\mathcal{S}(\mathrm{PG}(2, q); A, \ell, (X_s), \sigma)$ the $(q-1)(q-3)$ flags $(p, \delta(p))$, p a point of degree $q+1$, gives a $(q+2)$-regular incidence structure with q^2+q+6 points, q^2+q+6 lines and diameter 3.*

Proof. By Lemma 4.6 the new flags are not flags of $\mathcal{S}$ (the points of degree $q+1$ are off ℓ) and they match the elements of degree $q+1$ in pairs. Adding flags cannot increase distances, and $A \not| \ell$ still. □

Remark 4.8 (Arbitrary planes). In any projective plane of order $q \ge 7$ a regular completion also exists. Each of the $N = (q-1)(q-3)$ points of degree $q+1$ is incident with at most $q+1$ of the N lines of degree $q+1$, and $N - (q+1) \ge N/2$ for $q \ge 7$, so Hall's theorem gives a perfect matching of non-incident pairs.

Remark 4.9 (The three-hat structure of [2]). In the same coordinates, the structure of Araujo-Pardo and López uses the hats $S_{AL} = \{0, 1, a\}$, no rotation, and links A I K_a, H_a I ℓ, H_0 I K_1, H_1 I K_0; it removes only the flag (X_a, λ_a), and M contains P and X_t for $t \notin \{0, 1\}$, Q lies on λ_∞ and λ_t for $t \notin \{0, 1\}$. (In their affine coordinates $(x, y) = (x{:}y{:}1)$, $[m, b] = [m{:}{-}1{:}b]$, $L_c = [1{:}0{:}{-}c]$, $L = [0{:}0{:}1]$,

$P_m = (1{:}m{:}0)$; so $L_0 = \ell$, $P_0 = A$, $(0,s) = X_s$, $[0,s] = \lambda_s$ and $L = \lambda_\infty$.) It has $2(q^2+q+5)$ elements and diameter 3 for $q \geq 3$ [2]; the proof is a case analysis like that of Lemma 4.4. Its elements of degree $q+1$ are the points (x,y), $x \neq 0$, $y \notin S_{AL}$, and P_i, $i \neq 0$, and the lines $[x,y]$, $x \neq 0$, $y \notin S_{AL}$, and L_i, $i \neq 0$. The completing matching printed in [2] is $(x,y) \sim [x,y]$ together with $P_i \sim L_i$ for $i \in \mathbb{F} \setminus \{0,1,a\}$; with this index set the four elements P_1, P_a, L_1, L_a keep degree $q+1$. Taking $i \in \mathbb{F}^*$ instead (the pairs $P_i \notin L_i$ are always non-incident) gives a $(q+2)$-regular structure, as intended. This affects neither the maximum degree nor the diameter.

Remark 4.10 (Hats as doubled flags). Let $\mathcal{A} = \{\ell, A\} \cup \{X_s, \lambda_s : s \in S_{AL}\}$ in $\mathrm{PG}(2,q)$ and double it ($m = 2$), except that each flag (X_s, λ_s) is *crossed*: instead of $X'_s \mathrel{\mathrm{I}} \lambda'_s$ put $X'_s \mathrel{\mathrm{I}} \lambda_s$ and $X_s \mathrel{\mathrm{I}} \lambda'_s$. With $M = \ell'$, $Q = A'$, $H_s = X'_s$, $K_s = \lambda'_s$, one checks from the rules that the structure of [2] is this crossed doubling plus the six incidences $H_1 \mathrel{\mathrm{I}} K_0$, $H_0 \mathrel{\mathrm{I}} K_1$, $H_a \mathrel{\mathrm{I}} \ell$, $A \mathrel{\mathrm{I}} K_a$, $X_a \mathrel{\mathrm{I}} M$, $Q \mathrel{\mathrm{I}} \lambda_a$, minus the flag (X_a, λ_a). By Proposition 3.5 some patching is unavoidable: the component of $\mathcal{A}$ contains the path ℓ, X_s, λ_s, A of length $3 > n-2 = 1$. For $q = 5$ a direct check shows that in the crossed doubling without patches the pairs at distance greater than 3 are exactly (H_s, X_t), (K_s, λ_t) $(s \neq t)$, (H_s, Q) and (K_s, M), all of them copies of pairs joined inside $\mathcal{A}$ by a path of length $2 = n-1$ (such as X_s, ℓ, X_t or X_s, λ_s, A), as in the proof of Lemma 3.3; the patches repair exactly these. Heuristically, the fourth hat needs the rotation because M now misses three points of ℓ, and each hat point H_t, $t \in T$, must reach the two of them other than X_t with only two spare incidences.

Remark 4.11 (Scope). All proofs above use only the axioms of a projective plane, so Theorem 4.2 holds in non-Desarguesian planes as well. For $\Delta = q+2 \leq 16$ the theorem gives 192, 276, 376 at $\Delta = 11, 13, 15$ ($q = 9, 11, 13$), two more than [2]; at $\Delta = 7$ ($q = 5$) it gives 72, below the known value 80.

5. Diameter 4: generalized quadrangles

By Lemma 2.1, an incidence structure without isolated elements has diameter at most 4 exactly when, for every anti-flag (p, L), some point of L is collinear with p. Every generalized quadrangle has this property, with a unique such point.

5.1. **Doubling in quadrangles.** By Lemma 3.3, in a generalized quadrangle a set $\mathcal{A}_0$ satisfies (C_4) exactly when its components are single elements, flags or stars. The following is Delorme's construction [7, 8] in this language.

Example 5.1 (Delorme)**.** Let Σ be a spread of $\mathrm{W}(q)$, let $v \in \Sigma$, and let $\mathcal{P}(v)$ be the set of points of v. Put

$$\mathcal{A}_1 = \{v\}, \qquad \mathcal{A}_0 = \Sigma \cup \mathcal{P}(v).$$

(L1): the elements incident with v are its points; a point of v is incident with v only, a line of $\Sigma \setminus \{v\}$ with nothing in $\mathcal{A}_1$. (L0): a point off v is on exactly one line of Σ and on no other line of $\mathcal{A}_0$; a line not in Σ meets v in at most one point and contains no other element of $\mathcal{A}_0$. The components are the star $\{v\} \cup \mathcal{P}(v)$ and the single lines of $\Sigma \setminus \{v\}$. By Theorem 3.2 with $m = 2$ we get maximum degree $q+2$, diameter 4 and

$$2(q+1)(q^2+1) + |\mathcal{A}_0| + 2|\mathcal{A}_1| = 2(q+1)(q^2+1) + q^2 + q + 4 = 2q^3 + 3q^2 + 3q + 6$$

elements. These are the previous table values 346, 1734, 3064, 4946 at $\Delta = 7, 11, 13, 15$. Dually: an ovoid $\mathcal{O}$ of $\mathrm{Q}(4,q)$, a point $o \in \mathcal{O}$, $\mathcal{A}_1 = \{o\}$, $\mathcal{A}_0 = \mathcal{O} \cup \{\text{lines on } o\}$.

Remark 5.2 (Ovoids and stars). More generally, let $\mathcal{O}$ be an ovoid of $\mathrm{Q}(4,q)$ and $\mathcal{M}$ a set of lines such that every point outside $\mathcal{O}$ is on at most one line of $\mathcal{M}$. Then $\mathcal{A}_0 = \mathcal{O} \cup \mathcal{M}$ (one level) satisfies (L0) and (C_4): its components are stars centered at points of $\mathcal{O}$. This gives maximum degree $q+2$, diameter 4 and $2(q+1)(q^2+1) + q^2 + 1 + |\mathcal{M}|$ elements. If some $o \in \mathcal{O}$ is on $k \geq 2$ lines of $\mathcal{M}$, then $|\mathcal{M}| \leq k + (q+1-k)q \leq q^2 - q + 2$. Indeed, a line $L \in \mathcal{M}$ not on o has a unique point y

collinear with o; $y \notin \mathcal{O}$, the line oy is not in $\mathcal{M}$ (else y would be on two lines of $\mathcal{M}$), and distinct L give distinct y. If every point of $\mathcal{O}$ is on at most one line of $\mathcal{M}$, then $\mathcal{M}$ is a partial spread, so $|\mathcal{M}| \le q^2 - q + 1$ for q odd (Tallini [15][2]; for large partial ovoids see also [4]).

5.2. **Clones and shadow lines.** The next construction is not a doubling: it adjoins several copies of points and copies of lines that are attached to copies of points.

Definition 5.3 (Clones and shadows)**.** Let $\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$ be a generalized quadrangle of order (s,t). Let $w\colon \mathcal{P} \to \mathbb{Z}_{\ge 0}$ be a weight and $\mathcal{M}$ a finite multiset of lines. For each entry μ of $\mathcal{M}$, with underlying line L_μ, choose for every point $p \mathrel{\mathrm{I}} L_\mu$ a *stand-in* $\sigma_\mu(p) \in \{p, p^1, \dots, p^{w(p)}\}$. The structure $\mathcal{S}[w, \mathcal{M}, \sigma]$ has

- points: $\mathcal{P}$ and the *clones* $p^1, \dots, p^{w(p)}$ of each $p \in \mathcal{P}$;
- lines: $\mathcal{L}$ and one *shadow line* L^μ for each entry μ of $\mathcal{M}$;
- incidences: those of $\mathcal{S}$; $p^i \mathrel{\mathrm{I}} L$ whenever $p \mathrel{\mathrm{I}} L \in \mathcal{L}$; and $\sigma_\mu(p) \mathrel{\mathrm{I}} L^\mu$ for $p \mathrel{\mathrm{I}} L_\mu$.

So a clone is incident with exactly the old lines through its point, and a shadow line has exactly one point "over" each point of its line. Write π for the map sending a clone or a shadow line to the underlying point or line (π is the identity on $\mathcal{P} \cup \mathcal{L}$).

Theorem 5.4. *Let $\mathcal{S}$ be a generalized quadrangle of order (s,t), $s,t \ge 1$, and let $\mathcal{S}' = \mathcal{S}[w, \mathcal{M}, \sigma]$.*

(a) *$\mathcal{S}'$ has diameter 4.*

(b) *$\Delta(\mathcal{S}') \le \Delta$ if and only if $\Delta \ge \max(s,t) + 1$, $\sum_{p \mathrm{I} L} w(p) \le \Delta - s - 1$ for every line L of $\mathcal{S}$, and every point of $\mathcal{S}'$ (old point or clone) is incident with at most $\Delta - t - 1$ shadow lines.*

(c) *$\mathcal{S}'$ has $(s+1)(st+1) + \sum_p w(p)$ points and $(t+1)(st+1) + |\mathcal{M}|$ lines.*

Proof. (b) and (c) are counts: an old line L has $s + 1 + \sum_{p \mathrm{I} L} w(p)$ points, a shadow line $s+1$ points, and an old point or clone is on $t+1$ old lines plus the shadow lines it serves.

(a) We use Lemma 2.1(i). Let (x, Y) be an anti-flag of $\mathcal{S}'$. Every point x of $\mathcal{S}'$ is incident with every old line through $\pi(x)$. If Y is an old line, then $\pi(x) \not\mathrel{\mathrm{I}} Y$; let r be the point of Y collinear with $\pi(x)$ in $\mathcal{S}$, on the line N. Then $x \mathrel{\mathrm{I}} N \mathrel{\mathrm{I}} r \mathrel{\mathrm{I}} Y$. If $Y = L^\mu$ and $\pi(x) \not\mathrel{\mathrm{I}} L_\mu$, let $r \mathrel{\mathrm{I}} L_\mu$ be collinear with $\pi(x)$ on the line N. Then $x \mathrel{\mathrm{I}} N \mathrel{\mathrm{I}} \sigma_\mu(r) \mathrel{\mathrm{I}} L^\mu$. If $Y = L^\mu$ and $\pi(x) \mathrel{\mathrm{I}} L_\mu$, pick $r \mathrel{\mathrm{I}} L_\mu$ with $r \ne \pi(x)$. Then $x \mathrel{\mathrm{I}} L_\mu \mathrel{\mathrm{I}} \sigma_\mu(r) \mathrel{\mathrm{I}} L^\mu$. In each case some point of Y is collinear with x, so $\operatorname{diam} \mathcal{S}' \le 4$. Two non-collinear points p, p' of $\mathcal{S}$ remain non-collinear: they share no old line, and a shadow line L^μ contains only points over points of L_μ, and an old point is over itself only. So $\operatorname{diam} \mathcal{S}' = 4$. □

If all stand-ins are the old points, a shadow line is a duplicate of its line except that it misses the clones; this is the setting of the following corollary.

Corollary 5.5. *Let $\mathcal{S}$ be a generalized quadrangle of order (s,t) and $\Delta \ge \max(s,t) + 1$. Suppose $\mathcal{S}$ has a weighted $(\Delta - s - 1)$-ovoid w and a weighted $(\Delta - t - 1)$-cover $\mathcal{M}$. Then $\mathcal{S}[w, \mathcal{M}, \mathrm{id}]$ has diameter 4, maximum degree Δ and $2\Delta(st+1)$ elements. Adjoining further flags between clones and shadow lines makes it Δ-regular, with the same number of elements and diameter 4.*

Proof. Double counting flags gives $\sum_p w(p) = (\Delta - s - 1)(st+1)$ and $|\mathcal{M}| = (\Delta - t - 1)(st+1)$, and the total is $(s+t+2)(st+1) + (\Delta - s - 1)(st+1) + (\Delta - t - 1)(st+1) = 2\Delta(st+1)$. Old points and old lines have degree exactly Δ. Each clone has degree $t+1$ and lacks $\Delta - t - 1$; each shadow line has degree $s+1$ and lacks $\Delta - s - 1$; and $\sum_p w(p)\,(\Delta - t - 1) = |\mathcal{M}|\,(\Delta - s - 1)$. Order the clones $c_0, c_1, \dots$ and the shadow lines $y_0, \dots, y_{b-1}$, $b = |\mathcal{M}|$, and join c_i to y_j for $j \equiv i(\Delta - t - 1) + k \pmod b$, $0 \le k < \Delta - t - 1$. Since $\Delta - t - 1 \le b$ these are distinct, and every shadow line receives exactly $\Delta - s - 1$ of them. Adding flags cannot increase distances, and two non-collinear old points still have no common line. □

[2]Note for co-authors: the page range of this reference (Mitt. Math. Sem. Giessen 201 (1991) 141–147) is quoted from memory and has not been verified.

5.3. **Two elliptic quadrics in** $\mathrm{AS}(q)$**.** Let q be odd. Take $\mathrm{W}(q)$ with the alternating form $\beta(u,v) = u_0v_1 - u_1v_0 + u_2v_3 - u_3v_2$ and the point $x = (1{:}0{:}0{:}0)$. Then $x^\perp$ is the plane $X_1 = 0$, and the points of $\mathrm{AS}(q) = P(\mathrm{W}(q), x)$ are the q^3 points $(a{:}1{:}w_1{:}w_2)$, which we write (a, w) with $a \in \mathbb{F}$, $w = (w_1, w_2) \in \mathbb{F}^2$. The lines of $\mathrm{AS}(q)$ are:

- the *fibers* $\{(a_0 + \mu, w_0) : \mu \in \mathbb{F}\}$: the hyperbolic lines xy, $y \notin x^\perp$, with x removed;
- the sets $\{(a_0 + \lambda \det(w_0, d),\ w_0 + \lambda d) : \lambda \in \mathbb{F}\}$ with $d \neq 0$: the totally isotropic line through (a_0, w_0) and $(\det(w_0, d){:}0{:}d)$, with its point in $x^\perp$ removed.

Fix a non-square $\varepsilon \in \mathbb{F}$ and let $\mathrm{N}(w) = w_1^2 - \varepsilon w_2^2$, an anisotropic form. For $\kappa \in \mathbb{F}^*$ let

$$\mathcal{E}_\kappa\colon\ X_0X_1 = \kappa\,\mathrm{N}(X_2, X_3).$$

This is an elliptic quadric of $\mathrm{PG}(3, q)$ through x, with tangent plane $x^\perp$ at x. So $\mathcal{E}_\kappa \cap x^\perp = \{x\}$ and $\mathcal{E}_\kappa \setminus \{x\} = \{(\kappa\mathrm{N}(w), w) : w \in \mathbb{F}^2\}$ is a set of q^2 points of $\mathrm{AS}(q)$.

Theorem 5.6. *Let q be odd, ε a non-square and $\kappa_1\kappa_2 = 1/(4\varepsilon)$. Then every line of $\mathrm{AS}(q)$ meets $\mathcal{E}_{\kappa_1} \setminus \{x\}$ and $\mathcal{E}_{\kappa_2} \setminus \{x\}$ in exactly two points in total, counted with multiplicity. The two quadrics meet only in x and $e = (0, 0)$. Hence $w = \mathbf{1}_{\mathcal{E}_{\kappa_1}\setminus x} + \mathbf{1}_{\mathcal{E}_{\kappa_2}\setminus x}$ is a weighted 2-ovoid of $\mathrm{AS}(q)$ with $w(e) = 2$.*

Proof. Fibers. A fiber lies on a line of $\mathrm{PG}(3, q)$ through x not in the tangent plane $x^\perp$, so it meets each $\mathcal{E}_\kappa$ in one point besides x (in coordinates, at $\mu = \kappa\mathrm{N}(w_0) - a_0$).

The other lines. Write $\rho = \det(w_0, d)$ and $B = \mathrm{N}(w_0 + d) - \mathrm{N}(w_0) - \mathrm{N}(d)$. The point with parameter λ lies on $\mathcal{E}_\kappa$ if and only if

$$\kappa\mathrm{N}(d)\lambda^2 + (\kappa B - \rho)\lambda + \kappa\mathrm{N}(w_0) - a_0 = 0.$$

As $\mathrm{N}(d) \neq 0$ this is a quadratic with $1+\chi(\mathrm{disc}_\kappa)$ roots, where χ is the quadratic character ($\chi(0) = 0$) and disc_κ the discriminant. With $w_0 = (b_0, c_0)$, $d = (b_1, c_1)$ one has $B = 2(b_0b_1 - \varepsilon c_0c_1)$ and

$$B^2 - 4\mathrm{N}(w_0)\mathrm{N}(d) = 4\varepsilon(b_0c_1 - c_0b_1)^2 = 4\varepsilon\rho^2,$$

hence

$$\mathrm{disc}_\kappa = (\kappa B - \rho)^2 - 4\kappa\mathrm{N}(d)(\kappa\mathrm{N}(w_0) - a_0) = 4\kappa\mathrm{N}(d)\,(a_0 - a_\kappa^*), \qquad a_\kappa^* = -\frac{4\varepsilon\kappa\rho^2 - 2B\rho + \rho^2/\kappa}{4\mathrm{N}(d)}.$$

Now $a_{\kappa_1}^* - a_{\kappa_2}^* = -\frac{\rho^2}{4\mathrm{N}(d)}(\kappa_1 - \kappa_2)\big(4\varepsilon - \frac{1}{\kappa_1\kappa_2}\big) = 0$; call the common value a^*. The line meets the two quadrics in

$$2 + \chi\big(\mathrm{N}(d)(a_0 - a^*)\big)\big(\chi(\kappa_1) + \chi(\kappa_2)\big) = 2$$

points, because $\chi(\kappa_1)\chi(\kappa_2) = \chi(\varepsilon) = -1$. (A double root, $a_0 = a^*$, gives one point on each quadric.)

Intersection. $\kappa_1 \neq \kappa_2$, since $\kappa_1^2 = 1/(4\varepsilon)$ is impossible. So $\kappa_1\mathrm{N}(w) = \kappa_2\mathrm{N}(w)$ forces $w = 0$. □

Corollary 5.7. *For every odd prime power q, $\mathrm{AS}(q)[w, \emptyset]$ with w as in Theorem 5.6 is $(q+2)$-regular of diameter 4 with $2q^2(q + 2)$ elements. Hence $n_b(q + 2, 4) \geq 2q^2(q + 2)$.*

Proof. Theorem 5.4 with $(s, t) = (q - 1, q + 1)$ and $\mathcal{M} = \emptyset$: every line gains exactly two points, every clone is on the $q + 2$ lines of its point, and there are $q^3 + (q + 2)q^2 + 2q^2$ elements. □

At $q = 9, 11, 13$ this gives 1782, 3146, 5070. Since the structure is regular, no choice of clones and shadows on $\mathrm{AS}(q)$ does better at $\Delta = q + 2$. Weights are needed: for $q = 3, 5$, $\mathrm{AS}(q)$ has no 2-ovoid without repeated points.

5.4. **A Singer 2-ovoid of** $\mathrm{W}(q)$**.** Let q be odd, $\mathbb{F} = \mathrm{GF}(q) \subset K = \mathrm{GF}(q^2) \subset E = \mathrm{GF}(q^4)$, and g a primitive element of E. Model $\mathrm{PG}(3,q)$ as $E^*/\mathbb{F}^*$; the point $\langle g^i\rangle$ has index i modulo $(q+1)(q^2+1)$. Write $\bar{y} = y^{q^2}$, $k' = k^q$ for $k \in K$, $\mathrm{N}(y) = y\bar{y} \in K$ and $\mathrm{Tr} = \mathrm{Tr}_{K/\mathbb{F}}$. Let $d = g^{(q^2+1)/2}$, so $\bar{d} = -d$ and $h = d^2 = g^{q^2+1}$ generates K^*. The form

$$b(y,z) = \mathrm{Tr}_{E/\mathbb{F}}(d\, y\bar{z}) = \mathrm{Tr}\big(d(y\bar{z} - \bar{y}z)\big)$$

is alternating and non-degenerate; let $\mathrm{W}(q)$ be its symplectic quadrangle. Multiplication by g^{q+1} scales b by an element of $\mathbb{F}^*$, so it preserves $\mathrm{W}(q)$. Its orbits on points are the classes

$$C_a = \{\langle g^i\rangle : i \equiv a \pmod{q+1}\}, \qquad |C_a| = q^2+1,$$

and $\langle y\rangle \in C_a$ if and only if $\mathrm{N}(y) \in h^a\mathbb{F}^*$. These are Ebert's elliptic quadrics partitioning $\mathrm{PG}(3,q)$ [10].

Proposition 5.8. *Let q be odd.*

(a) *Every line of $\mathrm{PG}(3,q)$ meets each class C_a in at most two points.*
(b) *Every totally isotropic line of $\mathrm{W}(q)$ meets $C_a \cup C_{q-a}$ in exactly two points.*

So the $(q+1)/2$ sets $C_a \cup C_{q-a}$, $0 \le a \le (q-1)/2$, are 2-ovoids of $\mathrm{W}(q)$ partitioning its points; dually, they are 2-covers of $\mathrm{Q}(4,q)$.

Proof. Let $L = \langle y,z\rangle_{\mathbb{F}}$ and $u = sy + tz$. Then $\mathrm{N}(u) = As^2 + \beta st + Ct^2$ with $A = \mathrm{N}(y)$, $C = \mathrm{N}(z)$, $\beta = y\bar{z} + \bar{y}z$, all in K. Put $\theta = y\bar{z} - \bar{y}z$; as $\bar{\theta} = -\theta$, $\theta = dk_0$ with $k_0 \in K$, and

$$\mathcal{D} := \beta^2 - 4AC = \theta^2 = hk_0^2, \qquad b(y,z) = \mathrm{Tr}(hk_0).$$

Fix $\omega \in K^*$ with $\omega' = -\omega$. For $\mu \in K^*$ put $\nu = \omega/\mu$; then $\mathrm{Tr}(\nu k) = 0$ if and only if $k \in \mu\mathbb{F}$. So $\langle u\rangle$ lies in the class $\{\langle u\rangle : \mathrm{N}(u) \in \mu\mathbb{F}^*\}$ if and only if $(s{:}t)$ is a zero of

$$f_\nu(s,t) = \mathrm{Tr}(\nu A)s^2 + \mathrm{Tr}(\nu\beta)st + \mathrm{Tr}(\nu C)t^2.$$

(a) Suppose $f_\nu \equiv 0$. Then $A, \beta, C \in \mu\mathbb{F}$. If $\mathcal{D} \neq 0$, then $h = \mathcal{D}/k_0^2 \in \mu^2\mathbb{F}^*/k_0^2$; but $\mathbb{F}^* \subseteq (K^*)^2$, as $q+1$ is even, so h would be a square in K^*, which it is not. If $\mathcal{D} = 0$, then $\mathrm{N}(u) = \mu(\alpha_0 s^2 + \beta_0 st + \gamma_0 t^2)$ with $\beta_0^2 = 4\alpha_0\gamma_0$ and $\alpha_0 \neq 0$ (as $A \neq 0$), which vanishes at $(s{:}t) = (-\beta_0 : 2\alpha_0)$, impossible for $u \neq 0$. So $f_\nu \not\equiv 0$, and it has $1 + \chi(\mathrm{disc}(\nu)) \le 2$ zeros, $\mathrm{disc}(\nu) = \mathrm{Tr}(\nu\beta)^2 - 4\mathrm{Tr}(\nu A)\mathrm{Tr}(\nu C)$.

(b) Expanding, $\mathrm{disc}(\nu) = \mathrm{Tr}(\nu^2\mathcal{D}) + \nu\nu' P$ with $P = 2\beta\beta' - 4AC' - 4A'C \in \mathbb{F}$. If L is totally isotropic then $hk_0 \in \omega\mathbb{F}$, so $e := h\mathcal{D} = (hk_0)^2 \in \mathbb{F}$, and

$$\mathrm{Tr}(h^2\nu'^2\mathcal{D}) = e(h\nu'^2 + h'\nu^2) = hh'\,\mathrm{Tr}(\nu^2\mathcal{D}).$$

Hence $\mathrm{disc}(h\nu') = \mathrm{N}_{K/\mathbb{F}}(h)\,\mathrm{disc}(\nu)$. Since h generates K^*, $\mathrm{N}_{K/\mathbb{F}}(h)$ generates $\mathbb{F}^*$ and is a non-square, so $\chi(\mathrm{disc}(h\nu')) = -\chi(\mathrm{disc}(\nu))$. The class belonging to $h\nu'$ is $\omega/(h\nu') \equiv \mu'/h \equiv (h\mu)^{-1}$ modulo $\mathbb{F}^*$ (using $\mu^{q+1} \in \mathbb{F}^*$); in indices, $a \mapsto q - a$. Scaling ν by $\mathbb{F}^*$ multiplies the discriminant by a square. So L meets $C_a \cup C_{q-a}$ in $(1 + \chi(\mathrm{disc}(\nu))) + (1 - \chi(\mathrm{disc}(\nu))) = 2$ points. The pairing $a \leftrightarrow q - a$ has no fixed point since $q+1$ is even. □

Remark 5.9. For q odd, $\mathrm{W}(q)$ is known to have m-ovoids for every even m (in particular 2-ovoids) and $(q+1)/2$-ovoids [3]; see also [6] for m-ovoids of $\mathrm{W}(q)$. We have not checked whether the 2-ovoids $C_a \cup C_{q-a}$, or the partition they form, coincide with those constructions; the proof above is self-contained.[3] The proof only uses that d^2 is a non-square in K: replacing d by any δ_0 with $\bar{\delta}_0 = -\delta_0$ gives a symplectic form for which $C_a \cup C_b$ is a 2-ovoid when $h^{a+b}\delta_0^2 \in \mathbb{F}^*$.

[3]Note for co-authors: we have confirmed from the abstract of [3] only the existence of m-ovoids of $\mathrm{W}(q)$ for even m and of $(q+1)/2$-ovoids, q odd. Whether partitions into 2-ovoids occur there, and whether [6] has a Singer-group description, still has to be checked.

5.5. **All degrees.**

Corollary 5.10. *Let q be a prime power and $\Delta \geq q+1$. There is a Δ-regular incidence structure of diameter 4 with*

$$2\Delta(q^2+1) \text{ elements if } q \text{ is even or } \Delta - q - 1 \text{ is even}, \qquad 2\Delta q^2 \text{ elements otherwise.}$$

In particular $n_b(\Delta, 4) \geq 2q^2\Delta$ for every prime power $q \leq \Delta - 1$.

Proof. Apply Corollary 5.5. Let $j = \Delta - q - 1$. If q is even, $\mathrm{Q}(4,q)$ has an ovoid $\mathcal{O}$ and a spread Σ; take $w = j\mathbf{1}_{\mathcal{O}}$ and $\mathcal{M} = j\Sigma$. If q is odd and j is even, take $w = j\mathbf{1}_{\mathcal{O}}$ and $\mathcal{M}$ equal to $j/2$ copies of a 2-cover of $\mathrm{Q}(4,q)$ from Proposition 5.8. If q is odd and j is odd, then $\Delta \geq q+2$; in $\mathrm{AS}(q)$, of order $(q-1, q+1)$, take $(\Delta - q)/2$ copies of the weighted 2-ovoid of Theorem 5.6 and $\Delta - q - 2$ copies of the spread of fibers. □

At $q = 13$, $\Delta = 16$ this gives $n_b(16, 4) \geq 5440$.

5.6. **Limits of the method.**

Proposition 5.11. *Let q be odd, $\mathcal{O}$ an ovoid of $\mathrm{Q}(4,q)$, $w = \mathbf{1}_{\mathcal{O}}$, and let $\mathcal{S}' = \mathrm{Q}(4,q)[w, \mathcal{M}, \sigma]$ have maximum degree $q+2$. Then $|\mathcal{M}| \leq q^2 - q + 2$.*

Proof. By Theorem 5.4(b) every old point and every clone serves at most one shadow line. So a point outside $\mathcal{O}$ is on at most one entry of $\mathcal{M}$, and a point $o \in \mathcal{O}$ on at most two (one served by o, one by its clone). In particular $\mathcal{M}$ has no repeated line. If some $o \in \mathcal{O}$ is on two lines $L_1, L_2 \in \mathcal{M}$, then every other $L \in \mathcal{M}$ misses o, and its unique point y collinear with o is not in $\mathcal{O}$ (points of an ovoid are pairwise non-collinear). The line oy is not in $\mathcal{M}$, since y is on at most one line of $\mathcal{M}$, so oy is one of the $q-1$ lines on o other than L_1, L_2, and y is one of its q points other than o. Distinct L give distinct y, so $|\mathcal{M}| \leq 2 + q(q-1)$. Otherwise $\mathcal{M}$ is a partial spread of $\mathrm{Q}(4,q)$ and $|\mathcal{M}| \leq q^2 - q + 1$ (Tallini [15]). □

So these structures have at most $2(q+1)(q^2+1) + (q^2+1) + (q^2-q+2) = 2q^2(q+2) + q + 5$ elements; to beat Corollary 5.7 one needs $|\mathcal{M}| \geq q^2 - 2q - 2$. For $q = 5$ there is a set $\mathcal{M}$ of 21 lines with suitable stand-ins (an explicit example, checked by computer), giving $n_b(7, 4) \geq 359$; we do not know whether $q^2 - 2q - 2$ lines are possible for $q \geq 9$. In particular, $q^2 + 1$ shadow lines (which would give $2(q^2+1)(q+2)$ elements) are impossible for odd q.

Remark 5.12 (Weighted 3-ovoids). Weighted 3-ovoids of $\mathrm{W}(q)$, that is weighted 3-covers of $\mathrm{Q}(4,q)$, exist at least for $q = 3, 5, 7$ (found by computer; data in `research/p4/d4/sols/w_q3_j3.json`, `research/p4/d4/sols/w_q5_j3.json`, `research/p4/d4/sols/w3_q7.json`). For $q = 5$ there is even a 3-ovoid, and the $(q+1)/2$-ovoids of [3] are 3-ovoids exactly when $q = 5$. Together with 2-covers they would remove the case distinction in Corollary 5.10 for $\Delta \geq q + 4$.

Question 5.13. Does $\mathrm{W}(q)$ have a weighted 3-ovoid for every odd q? Does $\mathrm{AS}(q)$ have a weighted 3-ovoid for every odd q (known for $q = 5, 7$ by computer search; logs `research/gq4x/runs/ilp_q5_m3_exact_mult3.log` and `research/gq4x/runs/ilp_q7_m3_exact_mult3.log`)?

6. Diameter 6: the split Cayley hexagon

Throughout this section q is odd (even q is treated in Section 7), $\mathbb{F} = \mathrm{GF}(q)$, and $\mathrm{H}(q)$ is the split Cayley hexagon on the quadric $\mathrm{Q}(6,q)$ of equation $X_0X_4 + X_1X_5 + X_2X_6 = X_3^2$, as in Section 2. Distances are taken in the incidence graph of $\mathrm{H}(q)$; two lines are opposite when they are at distance 6, and a point is at distance 1, 3 or 5 from a line.

6.1. **The frame.** Let $v = \langle e_1, e_6 \rangle$. It is a line of $\mathrm{H}(q)$: all six Tits equations involve no Grassmann coordinate p_{16}, and e_1, e_6 span a totally singular line. Fix $c \in \mathbb{F}$ with $-c$ a non-square. For $z = (\alpha, \beta) \in \mathbb{F}^2$ let

$$\Pi_z\colon\ X_0 + cX_4 + \alpha X_2 + \beta X_5 = 0, \qquad \Pi = \Pi_{(0,0)},$$

and let S_z be the set of lines of $\mathrm{H}(q)$ contained in Π_z, $S = S_{(0,0)}$. All Π_z contain v.

Lemma 6.1. *For every $z \in \mathbb{F}^2$:*

(a) $\Pi_z \cap \mathrm{Q}(6,q)$ *is an elliptic quadric* $\mathrm{Q}^-(5,q)$.
(b) *Every point of* $\mathrm{H}(q)$ *in* Π_z *lies on exactly one line of* S_z. *So* $|S_z| = q^3+1$, *the lines of* S_z *are pairwise opposite, and* $v \in S_z$. (S_z *is a Hermitian spread.*)
(c) *A line of* $\mathrm{H}(q)$ *not in* S_z *meets* Π_z *in exactly one point.*
(d) *The points of* $\mathrm{H}(q)$ *at distance at most* 3 *from* v *are exactly the points of* $\mathrm{Q}(6,q)$ *with* $X_2 = X_5 = 0$. *Among them only the points of* v *lie in* Π_z.
(e) $S_z \cap S_{z'} = \{v\}$ *for* $z \neq z'$.

Proof. (a) On Π_z substitute $X_0 = -cX_4 - \alpha X_2 - \beta X_5$:

$$X_0X_4 + X_1X_5 + X_2X_6 - X_3^2 = (X_1 - \beta X_4)X_5 + X_2(X_6 - \alpha X_4) - (X_3^2 + cX_4^2).$$

This is the sum of two hyperbolic planes and the binary form $X_3^2 + cX_4^2$, which is anisotropic because $-c$ is a non-square. So the form on Π_z is non-degenerate of Witt index 2.

(b) Let $y \in \Pi_z \cap \mathrm{Q}(6,q)$. The lines of $\mathrm{H}(q)$ through y are the lines through y of the totally singular plane π_y. By (a), $\Pi_z \cap \mathrm{Q}(6,q)$ contains no totally singular plane, so $\pi_y \not\subseteq \Pi_z$ and $\pi_y \cap \Pi_z$ is a single line through y; it is the only line of $\mathrm{H}(q)$ on y inside Π_z. As $\mathrm{Q}^-(5,q)$ has $(q+1)(q^3+1)$ points, $|S_z| = q^3+1$. Two lines of S_z are not concurrent by uniqueness. If two lines of S_z were at distance 4, a line N of $\mathrm{H}(q)$ would meet both in distinct points; then $N \subseteq \Pi_z$, so $N \in S_z$ is concurrent with them, a contradiction. So they are opposite.

(c) A line not contained in the hyperplane Π_z meets it in one point.

(d) Let $y = ae_1 + be_6$ be a point of v and let u be a point with yu a line of $\mathrm{H}(q)$. The Tits equations $p_{12} = p_{34}$ and $p_{65} = p_{30}$ and the polarity condition give $au_2 = 0$, $bu_5 = 0$ and $au_5 + bu_2 = 0$, hence $u_2 = u_5 = 0$. So the points at distance at most 3 from v satisfy $X_2 = X_5 = 0$. Conversely, $\mathrm{Q}(6,q) \cap \{X_2 = X_5 = 0\}$ is $X_0X_4 = X_3^2$, a cone with vertex v over a conic; it has $(q+1)+(q+1)q^2$ points, which is the number of points at distance at most 3 from v ($q+1$ on v, and q further points on each of the $q(q+1)$ other lines meeting v). If such a point also lies in Π_z, then $X_0 = -cX_4$ and $-cX_4^2 = X_3^2$, so $X_0 = X_3 = X_4 = 0$ and the point is on v.

(e) Let $m \in S_z \cap S_{z'}$, $m \neq v$. On m, $(\alpha - \alpha')X_2 + (\beta - \beta')X_5 = 0$, so the linear map $m \to \mathbb{F}^2$, $X \mapsto (X_2, X_5)$, has a non-trivial kernel, and m has a point with $X_2 = X_5 = 0$. By (d) that point is at distance at most 3 from v. But m and v are opposite by (b), so every point of m is at distance 5 from v. □

6.2. **Delorme's doubling and whiskers.** Let $\mathcal{P}(v)$ be the set of points of v, $\mathcal{L}_v$ the set of lines $\neq v$ meeting v, and $\mathcal{P}_\Pi$ the set of points of $\mathrm{H}(q)$ in Π (the points on lines of S). A point is *free* if it is not in Π and is at distance 5 from v. By Lemma 6.1(d) the points of Π other than those of v are at distance 5 from v, so there are

$$(q+1)(q^4+q^2+1) - (q+1) - q^2(q+1) - q^3(q+1) = q^3(q^2-1)$$

free points.

Definition 6.2 (Whiskers)**.** A *whisker set* is a set W of lines of $\mathrm{H}(q)$, each opposite v and not in S, such that no free point is on two lines of W. It is *exact* if every free point is on exactly one line of W.

A line m opposite v and not in S meets Π in one point, which lies on a unique line of $S \setminus \{v\}$, and its other q points are free. So $|W| \le q^3(q^2-1)/q = q^2(q^2-1)$, with equality if and only if W is exact.

Theorem 6.3. *Let W be a whisker set and $m \ge 2$. Put*

$$\mathcal{A}_2 = \{v\}, \qquad \mathcal{A}_1 = \{v\} \cup \mathcal{P}(v) \cup S, \qquad \mathcal{A}_0 = \mathcal{A}_1 \cup \mathcal{P}_\Pi \cup \mathcal{L}_v \cup W.$$

Then $\mathcal{A}$ satisfies (L0), (L1) *and* (C_6). *Hence* $\mathrm{H}(q)^{(m)}_{\mathcal{A}}$ *has maximum degree $q+m$, diameter* 6, *and*

$$2(q+1)(q^4+q^2+1) + (m-1)\big(q^4+2q^3+q^2+2q+2+|W|+m(q^3+q+2)+m^2\big)$$

elements.

For $W = \emptyset$ this is Delorme's construction [7, 8]; at $m = 2$ it gives $2(q+1)(q^4+q^2+1)+q^4+4q^3+q^2+4q+10$ elements, the previous table values 8992, 142464, 374452, 842048 at $\Delta = 7, 11, 13, 15$, and at $q = 13$, $m = 3$ the value 884062 at $\Delta = 16$.

Proof. (L1), $i = 1$: the elements incident with v are its points, which are in $\mathcal{A}_1$; a point of v is incident with exactly one element of $\mathcal{A}_2$, a line of $S \setminus \{v\}$ with none. (L1), $i = 0$: the elements incident with $\mathcal{A}_1$ are the points of v, the lines of $\mathcal{L}_v$, and the points on lines of S, all in $\mathcal{A}_0$. A line of $\mathcal{L}_v$ meets v in one point and contains no other point of $\mathcal{A}_1$. A point of $\mathcal{P}_\Pi \setminus \mathcal{P}(v)$ is on exactly one line of S (Lemma 6.1(b)) and not on v. A line of W contains no point of v, being opposite v. (L0): a point outside $\mathcal{A}_0$ is a point off Π. If it is at distance 3 from v, it is on exactly one line of $\mathcal{L}_v$ (two would close a circuit of length at most 6) and on no line of W (that line would be at distance at most 4 from v). If it is free, it is on no line of $\mathcal{L}_v$ and on at most one line of W. A line outside $\mathcal{A}_0$ is not in Π, so it meets Π in one point, and that point is its only point in $\mathcal{A}_0$. (C_6): a line of $\mathcal{L}_v$ contains no point of Π other than its point on v (Lemma 6.1(d)), and a line of W contains exactly one point of Π, not on v. So the components of the substructure on $\mathcal{A}_0$ are

$$\{v\} \cup \mathcal{P}(v) \cup \mathcal{L}_v \qquad \text{and} \qquad \{n\} \cup \mathcal{P}(n) \cup \{w \in W : w \text{ meets } n\} \quad (n \in S \setminus \{v\}).$$

Both are trees of diameter at most 4: the first is the ball of radius 2 about v, the second consists of n, its points, and whiskers hanging from them. Lemma 3.3 gives (C_6). Now Theorem 3.2 applies. The diameter is exactly 6 by (1), because there are opposite elements of level -1: a free point p (level -1) has q^5 opposite points, and only $(q+1)(q^3+1) < q^5$ points lie in $\mathcal{A}_0$. For the count: $\mathcal{A}_2 = \{v\}$; $\mathcal{A}_1$ consists of q^3+1 lines and $q+1$ points, q^3+q+2 elements; and $\mathcal{A}_0$ consists of the q^3+1 lines of S, the q^2+q lines of $\mathcal{L}_v$, the $|W|$ whiskers and the $(q+1)(q^3+1)$ points of Π, which is $q^4+2q^3+q^2+2q+2+|W|$ elements. □

6.3. **An exact whisker set for every odd q.** For a point X of $\mathrm{H}(q)$ in Π, not on v, put

$$\nu(X) = \frac{X_2X_3 - X_4X_5}{cX_2^2 + X_5^2}.$$

This is well defined: the denominator vanishes only when $X_2 = X_5 = 0$ (as $-c$ is a non-square), and by Lemma 6.1(d) such a point of Π lies on v. Let

$$G(a,b) = (c_0a + c_1b)(a^2 + cb^2) - b^3, \qquad F(z) = \frac{G(\alpha,\beta)}{\alpha^2 + c\beta^2} \quad (z = (\alpha,\beta) \ne 0),$$

where $c_0, c_1 \in \mathbb{F}$ are chosen so that G has no zero on $\mathbb{F}^2 \setminus \{0\}$.

Lemma 6.4. *Such c_0, c_1 exist for every odd q.*

Proof. $G(a,0) = c_0a^3$, so we need $c_0 \ne 0$. For $b \ne 0$ put $\alpha = a/b$: $G(a,b) = 0$ if and only if $c_0\alpha + c_1 = 1/(\alpha^2+c)$ (note $\alpha^2 + c \ne 0$). For fixed α this is a line in the (c_0,c_1)-plane with $q-1$ points having $c_0 \ne 0$. The lines for $\alpha = 0$ and $\alpha = 1$ meet at $c_0 = 1/(1+c) - 1/c = -1/(c(1+c)) \ne 0$ (note $c \ne -1$ as $-c$ is a non-square). So the q lines cover fewer than $q(q-1)$ of the points with $c_0 \ne 0$, and some (c_0,c_1) is good. □

Theorem 6.5. *Let q be odd, $-c$ a non-square, and c_0, c_1 as in Lemma 6.4. Then*

$$W = \{\, m \in S_z \setminus \{v\} \;:\; z \in \mathbb{F}^2 \setminus \{0\},\ \nu(m \cap \Pi) = F(z) \,\}$$

is an exact whisker set: every free point lies on exactly one line of W, and $|W| = q^2(q^2-1)$.

Here $m \cap \Pi$ is a single point not on v: $m \subseteq \Pi_z$, $m \neq v$, so $m \notin S$ by Lemma 6.1(e), and m is opposite v by Lemma 6.1(b).

The proof rests on a complex structure on each hyperplane Π_z and on an identity that it yields. We use the following notation. A point of Π_z, $z = (\alpha, \beta)$, is written $x = (x_1, \dots, x_6)$, meaning $(X_0, \dots, X_6) = (-cx_4 - \alpha x_2 - \beta x_5,\ x_1, \dots, x_6)$. On Π_z the quadratic form is

$$Q_z(x) = x_1x_5 + x_2x_6 - x_3^2 - cx_4^2 - \alpha x_2x_4 - \beta x_4x_5,$$

with polar form B_z. Put $\mathrm{num}(y) = y_2y_3 - y_4y_5$ and $\mathrm{den}(y) = cy_2^2 + y_5^2$, so that $\nu = \mathrm{num}/\mathrm{den}$ on Π. Define the linear map J_z of Π_z by

$$J_z x = \big(-\beta x_3 + \alpha x_4 - x_6,\ x_5,\ -cx_4 - \beta x_5,\ x_3 + \beta x_2,\ -cx_2,\ cx_1 + \alpha\beta x_2 + \alpha x_3 + c\beta x_4 + \beta^2 x_5\big);$$

for $z = 0$ this is $J_0 x = (-x_6, x_5, -cx_4, x_3, -cx_2, cx_1)$.

Lemma 6.6 (Complex structure)**.** *For every $z \in \mathbb{F}^2$:*

(a) *$J_z^2 = -c$, $Q_z(J_z x) = cQ_z(x)$ and $B_z(x, J_z x) = 0$.*
(b) *The six differences $p_{12} - p_{34}$, $p_{54} - p_{32}$, $p_{20} - p_{35}$, $p_{65} - p_{30}$, $p_{01} - p_{36}$, $p_{46} - p_{31}$ of the Grassmann coordinates of $x \wedge J_z x$ are equal to $Q_z(x)$ times 1, 0, 0, $-c$, α, $-\beta$ respectively.*
(c) *For every point x of $\mathrm{Q}(6,q) \cap \Pi_z$, the line $\langle x, J_z x\rangle$ is the line of S_z through x.*
(d) *$\mathrm{num}(J_0 x) = c\,\mathrm{num}(x)$, $\mathrm{den}(J_0 x) = c\,\mathrm{den}(x)$, and the polar forms of* num *and* den *vanish on $(x, J_0 x)$. Consequently ν is constant on every line of $S \setminus \{v\}$.*

Proof. (a), (b) are polynomial identities of degree 2 in x, checked by direct expansion (we also checked them by computer algebra). For $z = 0$, for instance, $Q_0(J_0x) = cx_2x_6 + cx_1x_5 - c^2x_4^2 - cx_3^2 = cQ_0(x)$, and $p_{12} = x_1x_5 + x_2x_6$, $p_{34} = x_3^2 + cx_4^2$, so $p_{12} - p_{34} = Q_0(x)$.

(c) As $-c$ is a non-square, J_z has no eigenvector, so x and $J_z x$ span a line. If x is on the quadric, then $Q_z(x) = 0$, hence by (a) every point of the line is on the quadric, and by (b) its Grassmann coordinates satisfy Tits' equations. So it is a line of $\mathrm{H}(q)$ in Π_z through x, and by Lemma 6.1(b) it is the line of S_z through x.

(d) The first assertions are direct: $\mathrm{num}(J_0x) = x_5(-cx_4) - x_3(-cx_2) = c\,\mathrm{num}(x)$, $\mathrm{den}(J_0x) = cx_5^2 + c^2x_2^2 = c\,\mathrm{den}(x)$, and the polar forms are $x_2y_3 + x_3y_2 - x_4y_5 - x_5y_4$ and $2cx_2y_2 + 2x_5y_5$, which vanish for $y = J_0x$. By (c) every line $n \in S \setminus \{v\}$ is $\{ax + bJ_0x\}$ for any point x of n, and $\mathrm{num}(ax + bJ_0x) = (a^2 + cb^2)\mathrm{num}(x)$, $\mathrm{den}(ax + bJ_0x) = (a^2 + cb^2)\mathrm{den}(x)$. The factor $a^2 + cb^2$ is non-zero because $-c$ is a non-square. So ν is constant on n. □

In particular the condition defining W depends only on the line of S that m meets: ν is a function on the Hermitian spread $S \setminus \{v\}$.

Let p be a free point. By Lemma 6.1(d), $(p_2, p_5) \neq (0,0)$, and since $p \notin \Pi$ we may normalize $p_0 + cp_4 = 1$. Then $p \in \Pi_z$ if and only if z lies on the affine line

$$A_p = \{(\alpha, \beta) : \alpha p_2 + \beta p_5 = -1\} \subseteq \mathbb{F}^2 \setminus \{0\},$$

which has q points and direction $(p_5, -p_2)$. For $z \in A_p$ let $m_p(z)$ be the line of S_z through p (Lemma 6.1(b)) and $X(z) = m_p(z) \cap \Pi$.

Lemma 6.7 (Key identity)**.** *For $z = (\alpha, \beta) \in A_p$,*

$$\Phi_p(z) := \nu(X(z)) + \frac{\beta^3}{\alpha^2 + c\beta^2} = \frac{p_2p_3 - p_4p_5 + \beta\, p_2^2}{cp_2^2 + p_5^2}.$$

In particular Φ_p is the restriction to A_p of an affine function, and its increment along $(p_5, -p_2)$ is $-p_2^3/(cp_2^2+p_5^2)$. Explicitly, if $p_5 \neq 0$ and $z = (sp_5,\ -1/p_5 - sp_2)$, then

$$\Phi_p(z) = -\frac{p_2^3 p_5\, s + p_2^2 - p_2 p_3 p_5 + p_4 p_5^2}{p_5(cp_2^2+p_5^2)};$$

if $p_5 = 0$ and $z = (-1/p_2,\ s)$, then $\Phi_p(z) = (p_2 s + p_3)/(cp_2)$.

Proof. Put $\Theta = cp_2^2 + p_5^2 \neq 0$ and $\gamma = -\alpha p_5 + c\beta p_2$. Write $p = (p_1, \dots, p_6)$ in the coordinates of Π_z; then $\alpha p_2 + \beta p_5 = -1$. For $y \in \Pi_z$ the quantity $\lambda(y) = y_0 + cy_4$ equals $-\alpha y_2 - \beta y_5$, and Π is the set where it vanishes. We have $\lambda(p) = 1$ and, by the formula for J_z, $\lambda(J_z p) = -\alpha p_5 + c\beta p_2 = \gamma$. By Lemma 6.6(c), $m_p(z) = \langle p, J_z p\rangle$, so $X(z) = \gamma p - J_z p$. Its coordinates are

$$X_2 = \gamma p_2 - p_5, \quad X_3 = \gamma p_3 + cp_4 + \beta p_5, \quad X_4 = \gamma p_4 - p_3 - \beta p_2, \quad X_5 = \gamma p_5 + cp_2.$$

Direct expansion gives

$$\operatorname{den}(X) = (\gamma^2 + c)\,\Theta, \qquad \operatorname{num}(X) = (\gamma^2+c)(p_2p_3 - p_4p_5) + \beta\big(2\gamma p_2 p_5 + cp_2^2 - p_5^2\big).$$

Solving $\alpha p_2 + \beta p_5 = -1$, $-\alpha p_5 + c\beta p_2 = \gamma$ (determinant Θ) gives $\alpha = -(cp_2 + \gamma p_5)/\Theta$, $\beta = (\gamma p_2 - p_5)/\Theta$, and therefore $\alpha^2 + c\beta^2 = (\gamma^2 + c)/\Theta$. Here $\gamma^2 + c \neq 0$ since $-c$ is a non-square (this also shows $\operatorname{den}(X) \neq 0$). Hence

$$\nu(X) = \frac{p_2p_3 - p_4p_5}{\Theta} + \frac{\beta\,(2\gamma p_2 p_5 + cp_2^2 - p_5^2)}{(\gamma^2+c)\Theta}, \qquad \frac{\beta^3}{\alpha^2 + c\beta^2} = \frac{\beta\,(\gamma p_2 - p_5)^2}{(\gamma^2+c)\Theta},$$

using $\beta^2\Theta^2 = (\gamma p_2 - p_5)^2$. The sum of the two numerators multiplying $\beta/((\gamma^2+c)\Theta)$ is $2\gamma p_2 p_5 + cp_2^2 - p_5^2 + \gamma^2 p_2^2 - 2\gamma p_2 p_5 + p_5^2 = (\gamma^2 + c)p_2^2$, which gives the stated formula. Along $(p_5, -p_2)$ the coordinate β changes by $-p_2$. The two explicit formulas follow by inserting $\beta = -1/p_5 - sp_2$ and $\beta = s$. □

Remark 6.8. The identities in Lemmas 6.6 and 6.7 are polynomial identities with integer coefficients, so they hold in every odd characteristic. We also verified them symbolically, and, for $q = 3, 5, 7$, the construction of W itself on all free points of $\mathrm{H}(q)$.

Proof of Theorem 6.5. Let p be a free point, normalized as above. The lines of $\bigcup_{z\neq 0} S_z \setminus \{v\}$ through p are exactly the lines $m_p(z)$, $z \in A_p$, and these q lines are distinct: a common line would lie in $S_z \cap S_{z'} = \{v\}$. Since $F(z) = c_0\alpha + c_1\beta - \beta^3/(\alpha^2 + c\beta^2)$, we have $m_p(z) \in W$ if and only if

$$\Phi_p(z) - (c_0\alpha + c_1\beta) = 0.$$

The left side is affine on A_p with increment

$$-\frac{p_2^3}{cp_2^2+p_5^2} - (c_0p_5 - c_1p_2) = -\frac{G(p_5,-p_2)}{cp_2^2+p_5^2} \neq 0$$

along $(p_5, -p_2)$, so it vanishes at exactly one of the q points of A_p. Hence p is on exactly one line of W. The lines of W are opposite v and not in S, so W is an exact whisker set. □

Corollary 6.9. *For every odd prime power q and every $m \geq 2$ (even q: see Corollary 7.6),*

$$n_b(q+m, 6) \geq 2(q+1)(q^4+q^2+1) + (m-1)\big(2q^4 + 2q^3 + 2q + 2 + m(q^3+q+2) + m^2\big).$$

In particular $n_b(q+2,6) \geq 2(q+1)(q^4+q^2+1) + 2q^4 + 4q^3 + 4q + 10$ and $n_b(q+3,6) \geq 2(q+1)(q^4+q^2+1) + 4q^4 + 10q^3 + 10q + 34$.

By Proposition 3.6 an exact whisker set is the most this family can give: with W exact, every element outside $\mathcal{A}_0$ is incident with exactly one element of $\mathcal{A}_0$, and equality $q|\mathcal{A}_0| + 2c = 2N$ holds ($c = q^3 + 1$ components).

6.4. **Regular versions.** The structure of Theorem 6.3 with W exact has equally many points and lines in $\mathcal{A}_0$, but more lines than points in $\mathcal{A}_1$ and $\mathcal{A}_2$; it has $(m-1)m(q^3-q+m)$ more lines than points. We remove repeated lines and then add flags.

Lemma 6.10 (Repeated lines)**.** *If two lines of an incidence structure have the same set of points, deleting one of them changes no distance between the remaining elements.*

Proof. A shortest path through the deleted line L can pass through its twin L' instead; it does not already contain L', since then it could be shortened. □

Lemma 6.11 (Completion)**.** *Let $\mathcal{S}$ be a finite incidence structure with $\Delta(\mathcal{S}) \le \Delta$, such that every element has degree at least $\Delta - r$, and $\sum_p(\Delta - \deg p) = \sum_L(\Delta - \deg L) =: T$. If $T > 2r\Delta$, there is a set J of anti-flags such that $(\mathcal{P}, \mathcal{L}, \mathrm{I} \cup J)$ is Δ-regular.*

Proof. Choose a multiset J of point–line pairs in which each element x occurs $\Delta - \deg x$ times (possible as the sums agree). Call an entry *bad* if it is a flag of $\mathcal{S}$ or a repeated pair, and choose J with the fewest bad entries. Suppose (p, L) is bad. In $\mathrm{I} \cup J$, the line L is joined to at most Δ points and each of them occurs in at most r entries of J; likewise for p. So at most $2r\Delta$ entries (p', L') of J have p' joined to L or L' joined to p, and some entry (p', L') has neither. Replacing $(p, L), (p', L')$ by $(p, L'), (p', L)$ removes a bad entry and creates none. So J has no bad entries. □

Theorem 6.12. *For every odd prime power q and every $m \ge 2$ (even q: see Corollary 7.6) there is a $(q+m)$-regular incidence structure of diameter at most 6 with*

$$2(q+1)\big(q^4+q^2+1+(m-1)(q^3+m+1)\big)$$

elements; for $m = 2$ this is $2(q+1)(q^4+q^3+q^2+4)$, and for $m = 3$ it is $2(q+1)(q^4+2q^3+q^2+9)$.

Proof. Start from $\mathrm{H}(q)^{(m)}_{\mathcal{A}}$ with W exact. For $n \in S \setminus \{v\}$ (level 1) and $a \in [m]$, the m versions $n^{a1}, \dots, n^{am}$ all have the point set $\{p^a : p \ \mathrm{I} \ n\}$; for $a, b \in [m]$ the versions $v^{ab1}, \dots, v^{abm}$ all have the point set $\{p^{ab} : p \ \mathrm{I} \ v\}$. So there are $mq^3 + m^2$ classes of m repeated lines. Delete $(m-1)m(q^3-q+m) \le (m-1)(mq^3+m^2)$ lines, at most $m-1$ from each class; by Lemma 6.10 no distance increases, and points and lines are now equal in number. Every point of Π is on exactly one line of S, so each point loses at most $m-1$ lines. In $\mathrm{H}(q)^{(m)}_{\mathcal{A}}$ every version has degree between $q+1$ and $q+m$; after the deletion this still holds, so every deficit is at most $r = m-1$. The deficit sum T is at least that of the $m|W| = mq^2(q^2-1)$ whisker versions, each of degree $q+1$, so $T \ge (m-1)mq^2(q^2-1) > 2(m-1)(q+m)$. Lemma 6.11 completes the structure; adding flags cannot increase distances. The number of points is $(q+1)(q^4+q^2+1)+(m-1)\big((q+1)(q^3+1)+m(q+1)\big)$, which gives the formula. □

At $q = 5, 9, 11, 13$ and $m = 2$, Theorem 6.12 gives 9348, 147500, 386328, 866068 elements, and at $q = 13$, $m = 3$ it gives 927724. For these parameters the number of elements exceeds $M_b(q+m, 5)$, so the diameter is exactly 6.

7. Even q

Theorems 5.6, 6.5 and 6.12 and Corollaries 5.7 and 6.9 were stated for odd q. Here $q = 2^e$. What is proved for even q:

- $D = 4$, *every even q.* Corollary 5.10 already covers all prime powers: for even q it uses an ovoid and a spread of $\mathrm{Q}(4, q)$ and gives $2\Delta(q^2+1)$ elements for every $\Delta \ge q+1$ (Section 7.1). Theorem 5.6 and Corollary 5.7 (the quadrangle $\mathrm{AS}(q)$) are for odd q only and are not needed.
- $D = 6$, *even $q \ge 4$.* An explicit exact whisker set (Theorem 7.5); hence Corollary 6.9 and Theorem 6.12 hold for these q (Corollary 7.6).

- $D = 6$, $q = 2$. The formula for the whisker set fails (Lemma 7.2). We use an exact whisker set of 12 lines found by computer (an exact cover of the free points by an integer program). So for $q = 2$ Corollary 6.9 and Theorem 6.12 are computer-assisted; the resulting graphs were also built and checked directly (Table 2).

Throughout, $\mathbb{F} = \mathrm{GF}(q)$, Tr is the absolute trace of $\mathbb{F}$, and we fix $\delta \in \mathbb{F}$ with $\mathrm{Tr}(\delta) = 1$, so that $t^2 + t + \delta$ is irreducible over $\mathbb{F}$ and the binary form $a^2 + ab + \delta b^2$ is anisotropic.

7.1. **Diameter 4.** For q even nothing new is needed: $\mathrm{Q}(4, q)$ has both ovoids and spreads [14, Ch. 3], and Corollary 5.10 already gives a Δ-regular structure with $2\Delta(q^2 + 1)$ elements for every $\Delta \geq q + 1$. In the language of doubling (Section 3): if $\mathcal{O}$ is an ovoid and Σ a spread, every line of Σ contains exactly one point of $\mathcal{O}$, so $\mathcal{A}_0 = \mathcal{O} \cup \Sigma$ (one level) has $q^2 + 1$ components, each a flag, and satisfies (L0) and (C_4). Doubling gives maximum degree $q + 2$, diameter 4 and $2(q + 1)(q^2 + 1) + 2(q^2 + 1) = 2(q^2 + 1)(q + 2)$ elements; the elements of $\mathcal{A}_0$ and their copies have degree $q + 1$, and adding the anti-flags (p, L') and (p', L) for each flag (p, L) of $\mathcal{A}_0$ makes the structure $(q+2)$-regular. The structure of Corollary 5.5 with $w = \mathbf{1}_{\mathcal{O}}$, $\mathcal{M} = \Sigma$ has the same order. Every element outside $\mathcal{A}_0$ is incident with exactly one element of $\mathcal{A}_0$, so the counting bound of Proposition 3.6 holds with equality for $q^2 + 1$ components.

At $\Delta = q+2$ this beats the value $2q^2(q+2)$ of Theorem 5.6 by $2(q+2)$, so for even q the quadrangle $\mathrm{AS}(q)$ is not needed. For $q = 2, 4, 8, 16$ these graphs (and the ones with $j = 2$ copies of $\mathcal{O}$ and of Σ) were built and checked; see Table 2. They do not improve Table 3: when $q \in \{2, 4, 8, 16\}$, $q + 1$ is a prime power, so $\Delta = q + 2$ is a Moore degree, and at $\Delta = q + 3$ the odd prime power $q + 1$ gives more. The first degrees at which an even q is the best choice in Corollary 5.10 are $\Delta = 34, 35, 36, 37$ ($q = 32$; for example $n_b(34, 4) \geq 2 \cdot 1025 \cdot 34 = 69700$, against 65416 from $q = 31$), outside the range of the table.

7.2. **Diameter 6: the frame in characteristic 2.** The split Cayley hexagon $\mathrm{H}(q)$ is defined for all q by the same quadric and the same six Tits equations; in characteristic 2 the signs disappear and the quadric $\mathrm{Q}(6, q)\colon X_0X_4 + X_1X_5 + X_2X_6 + X_3^2 = 0$ has a nucleus, the point e_3. The hyperplanes $X_0 + cX_4 + \cdots = 0$ of Section 6 contain e_3 and so meet $\mathrm{Q}(6, q)$ in a cone; instead we take, for $z = (\alpha, \beta) \in \mathbb{F}^2$,

$$\Pi_z\colon \ X_3 = X_0 + \delta X_4 + \alpha X_2 + \beta X_5, \qquad \Pi = \Pi_{(0,0)},$$

and S_z the set of lines of $\mathrm{H}(q)$ in Π_z. Again $v = \langle e_1, e_6 \rangle \subseteq \Pi_z$ for all z.

Lemma 7.1. *Lemma 6.1 holds verbatim for even q with these Π_z.*

Proof. (a) On Π_z,

$$X_0X_4 + X_1X_5 + X_2X_6 + X_3^2 = \left(X_0^2 + X_0X_4 + \delta^2X_4^2\right) + (X_1 + \beta^2X_5)X_5 + X_2(X_6 + \alpha^2X_2).$$

The first term is anisotropic since $\mathrm{Tr}(\delta^2) = \mathrm{Tr}(\delta) = 1$, and Π_z misses the nucleus e_3, so $\Pi_z \cap \mathrm{Q}(6, q)$ is a non-degenerate quadric of Witt index 2, an elliptic quadric $\mathrm{Q}^-(5, q)$. (b), (c) and (e) are proved as before. (d) The Tits equations give $X_2 = X_5 = 0$ exactly as before (only signs change); the cone $X_0X_4 = X_3^2$ meets Π_z where $X_0^2 + X_0X_4 + \delta^2X_4^2 = 0$, that is $X_0 = X_4 = 0$, hence $X_3 = 0$, on v. □

So Theorem 6.3 (Delorme's doubling with whiskers) holds for even q without change. Free points are defined as before, there are $q^3(q^2 - 1)$ of them, and an exact whisker set has $q^2(q^2 - 1)$ lines.

7.3. **An exact whisker set for even q.** For a point X of $\mathrm{H}(q)$ in Π, not on v, put

$$\nu(X) = \frac{X_2X_3 + X_2X_4 + X_4X_5}{\delta X_2^2 + X_2X_5 + X_5^2}.$$

The denominator is anisotropic, and vanishes only if $X_2 = X_5 = 0$, that is (Lemma 7.1) only on v. For $c_0, c_1 \in \mathbb{F}$ let

$$G(a,b) = (c_0 a + c_1 b)(a^2 + ab + \delta b^2) + b^3, \qquad F(z) = \frac{G(\alpha,\beta)}{\alpha^2 + \alpha\beta + \delta\beta^2} \quad (z = (\alpha,\beta) \neq 0).$$

Lemma 7.2. *There are $c_0, c_1 \in \mathbb{F}$ such that G has no zero on $\mathbb{F}^2 \setminus \{0\}$ if and only if $q \geq 4$.*

Proof. $G(a,0) = c_0 a^3$, so $c_0 \neq 0$ is needed. For $b \neq 0$ and $t = a/b$, $G(a,b) = 0$ if and only if $c_0 t + c_1 = r(t) := 1/(t^2 + t + \delta)$. For fixed t this is a line in the (c_0, c_1)-plane with $q - 1$ points having $c_0 \neq 0$. If $q \geq 4$, take $\omega \notin \{0, 1\}$; then $r(\omega) \neq r(0)$, so the lines for $t = 0$ and $t = \omega$ meet at $c_0 = (r(0) + r(\omega))/\omega \neq 0$, and the q lines cover fewer than $q(q-1)$ points with $c_0 \neq 0$. If $q = 2$, then $r(0) = r(1) = 1$ and the lines $c_1 = 1$, $c_0 + c_1 = 1$ contain both points with $c_0 = 1$. □

We use coordinates $x = (x_0, x_1, x_2, x_4, x_5, x_6)$ on Π_z, with $X_3 = x_0 + \delta x_4 + \alpha x_2 + \beta x_5$, and define the linear map J_z of Π_z by

$$\begin{aligned} J_z x = \big(&(\delta+1)x_0 + \delta\alpha x_2 + \delta^2 x_4 + (\alpha+\delta\beta)x_5, \quad \beta x_0 + x_1 + \alpha\beta x_2 + (\alpha+\delta\beta)x_4 + \beta^2 x_5 + x_6, \\ &x_2 + x_5, \quad x_0 + (\alpha+\beta)x_2 + \delta x_4 + \beta x_5, \quad \delta x_2, \quad (\alpha+\beta)x_0 + \delta x_1 + \alpha^2 x_2 + \delta\alpha x_4 + \alpha\beta x_5 \big). \end{aligned}$$

Lemma 7.3 (Complex structure, characteristic 2)**.** *For every $z \in \mathbb{F}^2$, with Q the quadratic form and B its polar form:*

(a) *$J_z^2 + J_z + \delta = 0$, $Q(J_z x) = \delta Q(x)$ and $B(x, J_z x) = Q(x)$.*
(b) *The six Tits differences for $x \wedge J_z x$ are $Q(x)$ times $1, 0, 0, \delta, \alpha, \beta$.*
(c) *For every point x of $\mathrm{Q}(6,q) \cap \Pi_z$ the line $\langle x, J_z x\rangle$ is the line of S_z on x.*
(d) *With* num $= X_2X_3 + X_2X_4 + X_4X_5$ *and* den $= \delta X_2^2 + X_2X_5 + X_5^2$ *on Π,*

$$\mathrm{num}(sx + tJ_0 x) = (s^2 + st + \delta t^2)\,\mathrm{num}(x),$$

and likewise for den*. So ν is constant on every line of $S \setminus \{v\}$.*

Proof. (a), (b), (d) are polynomial identities over $\mathbb{F}_2[\delta, \alpha, \beta, x]$, checked by direct expansion (and by computer algebra, `research/evenq/proof_J.py`). (c) follows from (a), (b) and Lemma 7.1(b) as in Lemma 6.6: J_z has no eigenvector because $t^2 + t + \delta$ is irreducible, and on a singular x the line $\langle x, J_z x\rangle$ is totally singular with Grassmann coordinates satisfying Tits' equations. In (d), $s^2 + st + \delta t^2 \neq 0$ for $(s,t) \neq 0$. □

Let p be a free point. Then $(p_2, p_5) \neq (0,0)$, and we normalize $p_0 + p_3 + \delta p_4 = 1$. Now $p \in \Pi_z$ if and only if z lies on the affine line $A_p\colon \alpha p_2 + \beta p_5 = 1$, which has direction (p_5, p_2). For $z \in A_p$ let $X(z)$ be the point of Π on the line of S_z through p.

Lemma 7.4 (Key identity, characteristic 2)**.** *For $z = (\alpha,\beta) \in A_p$,*

$$\Phi_p(z) := \nu(X(z)) + \frac{\beta^3}{\alpha^2 + \alpha\beta + \delta\beta^2} = \frac{p_2p_3 + p_2p_4 + p_4p_5 + \beta p_2^2}{\delta p_2^2 + p_2p_5 + p_5^2}.$$

So Φ_p is affine on A_p, with increment $p_2^3/(\delta p_2^2 + p_2p_5 + p_5^2)$ along (p_5, p_2).

Proof. Put $\Theta = \delta p_2^2 + p_2p_5 + p_5^2 \neq 0$ and $\lambda(y) = y_0 + y_3 + \delta y_4$, so that $\Pi = \{\lambda = 0\}$ and $\lambda = \alpha y_2 + \beta y_5$ on Π_z. Then $\lambda(p) = 1$ and $\gamma := \lambda(J_z p) = \alpha(p_2 + p_5) + \delta\beta p_2$. By Lemma 7.3(c) the line of S_z on p is $\langle p, J_z p\rangle$, so $X(z) = \gamma p + J_z p$ (then $\lambda(X) = 2\gamma = 0$). In particular $X_2 = (\gamma+1)p_2 + p_5$ and $X_5 = \delta p_2 + \gamma p_5$. Direct expansion gives

$$\mathrm{den}(X) = (\gamma^2 + \gamma + \delta)\,\Theta, \qquad \mathrm{num}(X) = (\gamma^2 + \gamma + \delta)(p_2p_3 + p_2p_4 + p_4p_5) + \beta\big((\gamma + \delta + 1)p_2^2 + p_5^2\big),$$

and $\gamma^2+\gamma+\delta\neq 0$. Solving $\alpha p_2+\beta p_5=1$, $\alpha(p_2+p_5)+\delta\beta p_2=\gamma$ (determinant Θ) gives $\alpha=(\delta p_2+\gamma p_5)/\Theta$, $\beta=((\gamma+1)p_2+p_5)/\Theta$, hence $\alpha^2+\alpha\beta+\delta\beta^2=(\gamma^2+\gamma+\delta)/\Theta$ and

$$\frac{\beta^3}{\alpha^2+\alpha\beta+\delta\beta^2}=\frac{\beta\big((\gamma+1)p_2+p_5\big)^2}{(\gamma^2+\gamma+\delta)\Theta}.$$

The β-terms add up to $\beta\big((\gamma+\delta+1)+(\gamma+1)^2\big)p_2^2/((\gamma^2+\gamma+\delta)\Theta)=\beta p_2^2/\Theta$. Along (p_5,p_2), β changes by p_2. □

Theorem 7.5. *Let $q\geq 4$ be even and c_0,c_1 as in Lemma 7.2. Then*

$$W=\big\{\,m\in S_z\setminus\{v\}\ :\ z\in\mathbb{F}^2\setminus\{0\},\ \nu(m\cap\Pi)=F(z)\,\big\}$$

is an exact whisker set, so $|W|=q^2(q^2-1)$. For $q=2$ the formula gives none, since no (c_0,c_1) is good (Lemma 7.2); but an exact whisker set of $12=q^2(q^2-1)$ lines exists, found by computer (data: `research/evenq/whiskers_raw_q2.json`, program `research/evenq/explore.py`, mode `raw`). So Theorem 6.3 has an exact W for every even q, with a computer-found W when $q=2$.

Proof. As in Theorem 6.5: the lines of $\bigcup_{z\neq 0}S_z\setminus\{v\}$ on a free point p are the q distinct lines $m_p(z)$, $z\in A_p$, and $m_p(z)\in W$ if and only if $\Phi_p(z)+c_0\alpha+c_1\beta=0$, because $F(z)=c_0\alpha+c_1\beta+\beta^3/(\alpha^2+\alpha\beta+\delta\beta^2)$. By Lemma 7.4 the left side is affine on A_p with increment

$$\frac{p_2^3}{\Theta}+c_0p_5+c_1p_2=\frac{G(p_5,p_2)}{\Theta}\neq 0$$

along (p_5,p_2), so it vanishes at exactly one point of A_p. □

Corollary 7.6. *Corollary 6.9 and Theorem 6.12 hold for every even $q\geq 4$, and for $q=2$ with the computer-found whisker set of Theorem 7.5. Hence they hold for every prime power q (for $q=2$ computer-assisted): for $m\geq 2$,*

$$n_b(q+m,6)\geq 2(q+1)(q^4+q^2+1)+(m-1)\big(2q^4+2q^3+2q+2+m(q^3+q+2)+m^2\big),$$

and there is a $(q+m)$-regular incidence structure of diameter at most 6 with $2(q+1)\big(q^4+q^2+1+(m-1)(q^3+m+1)\big)$ elements.

Proof. The first bound is Theorem 6.3 with Lemma 7.1 and an exact W (Theorem 7.5). The proof of Theorem 6.12 uses, besides Theorem 6.3, only the facts that W is exact, that each point of Π is on exactly one line of S (Lemma 7.1) and the inequality $T\geq(m-1)mq^2(q^2-1)>2(m-1)(q+m)$, which holds for $q=2$ as well ($12m>2m+4$). □

As for $D=4$, this does not change Table 3: for $q=2,4,8,16$ the degree $q+2$ is a Moore degree (H$(q+1)$ exists), and at $\Delta=q+3$ the odd $q+1$ gives more. Even q is the best choice first at $\Delta=34,\dots,37$ ($q=32$; e.g. $n_b(34,6)\geq 71502028$ against 63159250 from $q=31$, $m=3$).

Remark 7.7 (What changes in characteristic 2). The anisotropic form a^2+cb^2 ($-c$ a non-square) becomes $a^2+ab+\delta b^2$ (Tr$\delta=1$); $J^2=-c$ becomes $J^2+J+\delta=0$; the hyperplanes Π_z must avoid the nucleus; ν gains the term X_2X_4; and Lemma 6.4 needs $q\geq 4$. Equivalently, with $K=\mathbb{F}[\omega]$, $\omega^2=\omega+\delta$: $\nu(X)=\mathrm{Tr}_{K/\mathbb{F}}\big(\omega\,(X_3+X_4\omega)/(\delta X_2+X_5\omega)\big)$.

Remark 7.8 (Computer checks and data). All code, graphs and logs are in `research/evenq/`: `hexq.py` builds H(q) for every q; `frame.py` and `check_frame.py` check Lemma 7.1 for $q=2,4,8$; `rule.py` and `rule_all.py` build W from Theorem 7.5 and check that it is exact if and only if G has no zero, for $q=2,4,8$ (log `runs/rule_q16.log` for $q=16$); `proof_J.py` and `proof_J2.py` verify Lemmas 7.3 and 7.4 over $\mathbb{F}_2[\delta,\alpha,\beta,x]$; `build_graph.py` and `d4_even.py` build the $D=6$ and $D=4$ graphs (in `graphs/`, with logs in `runs/`). The exact diameter of every graph in Table 2 was computed by BFS from every vertex ($q=16$, $D=6$: 2 384 492 vertices, about 32 hours).

D	q	Δ	construction	order	check
4	2, 4, 8, 16	$q+2$	ovoid + spread, regular	40, 204, 1300, 9252	all VALID
4	2, 4, 8, 16	$q+3$	2 ovoids + 2 spreads, regular	50, 238, 1430, 9766	all VALID
6	2, 4, 8	$q+2$	Thm. 6.3 + exact W	208, 3524, 85180	all VALID
6	2, 4, 8	$q+3$	same, $m=3$	324, 4468, 96516	all VALID
6	2, 4, 8	$q+2$	regular (Thm. 6.12)	192, 3400, 84168	all VALID
6	2, 4, 8	$q+3$	regular, $m=3$	270, 4090, 93474	all VALID
6	16	18	Thm. 6.3 + exact W	2384492	VALID

TABLE 2. Even q, computer checks (bipartite, maximum degree, exact diameter by BFS from every vertex). For $D=6$, $q\geq 4$ uses the whiskers of Theorem 7.5; $q=2$ uses an ILP solution. Exactness of W was also checked directly for $q=4$ ($\delta=2,3$), $q=8$ ($\delta=1,3,5,7$) and $q=16$ ($\delta=8,9$; 65280 whiskers). At each Δ above, the value is below the Moore bound where a generalized polygon of order $(\Delta-1,\Delta-1)$ exists, and below the corresponding entry of Table 3 otherwise.

8. Orders

At $\Delta=q+2$ the gap between the Moore bound and a generalized D-gon of order (q,q) is

$$M_b(q+2,D)-M_b(q+1,D)=2\sum_{i=1}^{D-1}\big((q+1)^i-q^i\big)=2(D-1)q^{D-2}+O(q^{D-3});$$

it is $4q+4$ for $D=3$ and $6q^2+10q+6$ for $D=4$. Doubling cannot close this gap. By Proposition 3.6, a doubled set satisfies $|\mathcal{A}_0|<2N/q$, where $N=q^{D-1}+O(q^{D-2})$ is the number of points; the higher levels are smaller still, since every element of $\mathcal{A}_1$ has all its $q+1$ neighbors in $\mathcal{A}_0$. So doubling gains at most $2q^{D-2}+O(q^{D-3})$ elements, roughly a fraction $1/(D-1)$ of the gap.

- $D=3$: the gain is a constant. A pure doubling gains at most 2 (Proposition 3.5); with patches, three hats gain 8 [2] and four hats gain 10 (Theorem 4.2).
- $D=4$: the gain is of order q^2. Delorme's doubling gains q^2+q+4. For even q, doubling an ovoid and a spread of $\mathrm{Q}(4,q)$ gains $2(q^2+1)$ and meets the counting bound of Proposition 3.6 with equality (Section 7.1). For odd q, Theorem 5.6 uses a different quadrangle, $\mathrm{AS}(q)$, and has $2q^2(q+2)=2(q+1)(q^2+1)+2q^2-2q-2$ elements, which has the order of magnitude of the counting bound $2q^2+O(q)$; Proposition 5.11 limits the shadow-line variant on $\mathrm{Q}(4,q)$ to at most $q+5$ more.
- $D=6$: the gain is of order q^4. Delorme's doubling gains $q^4+O(q^3)$; with an exact whisker set (Theorem 6.5 for odd q, Theorem 7.5 for even $q\geq 4$, a computer-found set for $q=2$) the gain is $2q^4+4q^3+4q+10$, and the single-level part attains the counting bound with equality.

Table 3 lists the resulting values in the range $\Delta\leq 16$ of the table of largest known bipartite graphs [5], for the degrees where no generalized polygon of order $(\Delta-1,\Delta-1)$ is known.

9. Open questions

(1) *Diameter* 3. Is there a five-hat structure, or any modification of a projective plane of order q with $\Delta=q+2$, diameter 3 and more than $2(q^2+q+6)$ elements?
(2) *Diameter* 4. For odd $q\geq 9$, are there q^2-2q-2 shadow lines on $\mathrm{Q}(4,q)$ with one ovoid of clones (Proposition 5.11)? Do weighted 3-ovoids of $\mathrm{W}(q)$ exist for all odd q? Do the 2-ovoids of Proposition 5.8 coincide with known ones [3, 6]?
(3) *Diameter* 6. Is there a balanced doubling of $\mathrm{H}(q)$, with level-1 centers at points as well as lines, that is regular without deleting repeated lines? The Feit–Higman theorem rules out hexagons

(Δ, D)	q	$M_b(\Delta, D)$	table [5]	new	regular	construction
$(11, 3)$	9	222	190	**192**	192	Thm. 4.2, 4.7
$(13, 3)$	11	314	274	**276**	276	Thm. 4.2, 4.7
$(15, 3)$	13	422	374	**376**	376	Thm. 4.2, 4.7
$(7, 4)$	5	518	346	**359**	350†	Thm. 5.4, Q$(4, 5)$, 21 shadow lines*
$(11, 4)$	9	2222	1734	**1782**	1782	Cor. 5.7, AS(9)
$(13, 4)$	11	3770	3064	**3146**	3146	Cor. 5.7, AS(11)
$(15, 4)$	13	5910	4946	**5070**	5070	Cor. 5.7, AS(13)
$(16, 4)$	13	7232	5134	**5440**	5440	Cor. 5.10, Q$(4, 13)$, Prop. 5.8
$(7, 6)$	5	18662	8992	**9592**	9348	Cor. 6.9, Thm. 6.12
$(11, 6)$	9	222222	142464	**148944**	147500	Cor. 6.9, Thm. 6.12
$(13, 6)$	11	542906	374452	**388972**	386328	Cor. 6.9, Thm. 6.12
$(15, 6)$	13	1158390	842048	**870440**	866068	Cor. 6.9, Thm. 6.12
$(16, 6)$	13	1627232	884062	**940846**	927724	Cor. 6.9, Thm. 6.12, $m = 3$

TABLE 3. Values for $\Delta \leq 16$ with $\Delta - 1$ not a prime power. "Table" is the value listed in [5] (consulted October 2026; for $D = 3$ these are the values of [2], for $D = 4$ and $D = 6$ those of Delorme [7, 8]). "New" is the largest structure in this paper; "regular" the largest Δ-regular one. *An explicit example, not a member of a proved infinite family (data: `research/gq4x/graphs/q4_share_q5_n359.json`). †Corollary 5.7 at $q = 5$ (data: `research/gq4x/graphs/as_w2_q5_n350.json`). The cells $(7, 3)$ and $(16, 3)$ are not improved: Theorem 4.2 gives $72 < 80$ at $(7, 3)$. All structures in the table were also built explicitly and their incidence graphs checked by computer. Every row uses an odd q. For $\Delta \leq 16$ the even prime powers $q \in \{2, 4, 8\}$ give smaller values in both the "new" and the "regular" columns for $D = 4$ and $D = 6$ (Section 7), so the table is unchanged.

of order $(q - 1, q + 1)$, so there is no direct analogue of AS(q); are there weighted distance-2 ovoids of H(q) that help?

(4) *Diameter* 8. The Ree–Tits octagon of order $(2, 4)$ has the shape $(q - 1, q + 1)$ with $q = 3$; a weighted 2-ovoid in it would give a 5-regular structure of diameter 8 with 5850 elements.

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